

# Traceability Matrix

REMEMBER GORDON! -- Rulebook  $\rightleftharpoons$  Implementation Mapping

Generated from `docs/traceability.toml`

## Overview

| Implemented | Descriptive | Implicit | Out-of-scope |
|-------------|-------------|----------|--------------|
| 76          | 10          | 6        | 16           |

Total mappings: 108 · Total impl sites: 224

## Table of Contents

|                                                                                |    |
|--------------------------------------------------------------------------------|----|
| Overview .....                                                                 | 2  |
| §1 -- Introduction .....                                                       | 7  |
| §1.1 -- General Comments .....                                                 | 7  |
| §1.2 -- Game Scale .....                                                       | 7  |
| §2 -- Game Components .....                                                    | 8  |
| §2.1 -- The Game Maps .....                                                    | 8  |
| §2.2 -- Play Aids .....                                                        | 8  |
| §2.3 -- The Units .....                                                        | 8  |
| §2.4 -- Game Parts Inventory .....                                             | 9  |
| §2.31 -- Dervish weapon types .....                                            | 9  |
| §2.32 -- Anglo-Egyptian weapon types .....                                     | 10 |
| §3 -- Getting Started .....                                                    | 11 |
| §3 -- Getting Started .....                                                    | 11 |
| §4 -- Turn Sequence .....                                                      | 12 |
| §4 -- Turn Sequence .....                                                      | 12 |
| §5 -- Movement Phase .....                                                     | 14 |
| §5 -- Movement Phase (general) .....                                           | 14 |
| §5.3 -- Constructing the Zariba .....                                          | 14 |
| §5.11 -- Movement allowances printed on units .....                            | 15 |
| §5.12 -- Move up to allowance, hex by hex (cumulative MP cap) .....            | 16 |
| §5.13 -- No MP accumulation between turns .....                                | 16 |
| §5.21 -- Friendlies transport via gunboat .....                                | 17 |
| §5.22 -- Land units may never enter a Nile River hex .....                     | 18 |
| §5.23 -- Walled city entry restrictions .....                                  | 18 |
| §5.24 -- Gunboat upstream/downstream movement .....                            | 19 |
| §5.25 -- Dervish forts may not move .....                                      | 19 |
| §5.26 -- Units stop on entering enemy ZOC .....                                | 20 |
| §5.41 -- All units except AE leaders exert ZOC .....                           | 20 |
| §5.42 -- No MP cost to enter/leave enemy ZOC .....                             | 20 |
| §5.43 -- Units stop when entering enemy ZOC .....                              | 21 |
| §5.44 -- ZOC limitations (walls, khor, fort, Nile, Zariba) .....               | 21 |
| §5.51 -- Stacking limit (4 units + leaders, gunboats isolated) .....           | 22 |
| §5.52 -- Different Dervish tribes may not stack together .....                 | 22 |
| §5.53 -- Leader stacking with command colour only .....                        | 23 |
| §5.54 -- Anglo-Egyptian Brigade Integrity .....                                | 23 |
| §6 -- Fire Combat Phase .....                                                  | 25 |
| §6.3 -- Line of Sight Table .....                                              | 25 |
| §6.6 -- Special Artillery Capabilities .....                                   | 26 |
| §6.7 -- Defensive Fire .....                                                   | 26 |
| §6.11 -- Fire combat factor printed on units .....                             | 26 |
| §6.12 -- Fire combat is always voluntary .....                                 | 27 |
| §6.13 -- Fire factor is unitary (may not be divided) .....                     | 27 |
| §6.14 -- Players may combine fire factors into one attack .....                | 27 |
| §6.15 -- May divide a stack to fire at different hexes .....                   | 27 |
| §6.16 -- Halving fire strength rounds down, minimum 1 .....                    | 27 |
| §6.21 -- First check LOS before firing .....                                   | 28 |
| §6.22 -- Consult Range Effects Table .....                                     | 28 |
| §6.23 -- Terrain defensive modifier .....                                      | 29 |
| §6.24 -- Anglo-Egyptian direct fire accuracy bonus and brigade integrity ..... | 30 |
| §6.41 -- Direct Fire Subphase .....                                            | 30 |
| §6.42 -- Maxim Second Fire and Howitzer Fire Subphase .....                    | 30 |
| §6.51 -- Leader Units .....                                                    | 31 |
| §6.52 -- Anglo-Egyptian Friendlies Brigade .....                               | 31 |
| §6.53 -- Royal Engineers demolition .....                                      | 31 |
| §6.54 -- Forts .....                                                           | 32 |
| §6.61 -- Only artillery may fire at gunboats; 3+ to sink .....                 | 33 |
| §6.62 -- Only artillery may fire at forts; 2+ to destroy .....                 | 33 |
| §6.63 -- Only artillery may breach wall hexsides; 2+ required .....            | 34 |
| §6.64 -- Howitzer fire .....                                                   | 34 |
| §6.81 -- Moving player may fire with all capable units .....                   | 35 |

|                                                                          |    |
|--------------------------------------------------------------------------|----|
| §6.82 -- Advance after combat (offensive fire) .....                     | 35 |
| §7 -- Melee Phase .....                                                  | 37 |
| §7.1 -- Melee strength printed on counter .....                          | 37 |
| §7.2 -- Melee adjacent only, not across wall hexsides .....              | 37 |
| §7.3 -- Simultaneous melee combat .....                                  | 38 |
| §7.4 -- Who may melee attack / defend .....                              | 38 |
| §7.5 -- Cavalry/camel retreat before melee .....                         | 39 |
| §7.6 -- Advance after melee .....                                        | 39 |
| §7.7 -- Melee modifiers .....                                            | 40 |
| §8 -- Night Game Turns .....                                             | 42 |
| §8.1 -- Night effects .....                                              | 42 |
| §8.2 -- Dervish Desertion Roll .....                                     | 42 |
| §9 -- The Scenarios .....                                                | 44 |
| §9.1 -- The Campaign Game .....                                          | 44 |
| §9.2 -- The Historical Scenario .....                                    | 44 |
| §9.3 -- Bonus Game: Fall of Khartoum .....                               | 44 |
| §9.11 -- Set Up (Campaign) .....                                         | 44 |
| §9.12 -- Scenario Length (Campaign) .....                                | 44 |
| §9.13 -- Special Rules (Campaign) .....                                  | 45 |
| §9.14 -- Victory Conditions (Campaign) .....                             | 45 |
| §9.21 -- Set Up (Historical) .....                                       | 47 |
| §9.22 -- Scenario Length (Historical) .....                              | 47 |
| §9.23 -- Special Rule: The Zariba .....                                  | 47 |
| §9.24 -- Victory Conditions (Historical) .....                           | 48 |
| §9.31 -- Bonus game map .....                                            | 48 |
| §9.32 -- Set Up (Bonus) .....                                            | 49 |
| §9.33 -- Scenario Length (Bonus) .....                                   | 49 |
| §9.34 -- Special Rules (Bonus) .....                                     | 49 |
| §9.35 -- Victory Conditions (Bonus) .....                                | 49 |
| §9.111 -- Dervish set up (Campaign) .....                                | 50 |
| §9.112 -- Dervish reinforcements (Campaign) .....                        | 50 |
| §9.113 -- Anglo-Egyptian set up (Campaign) .....                         | 51 |
| §9.211 -- Anglo-Egyptian set up first, moves second (Historical) .....   | 51 |
| §9.212 -- Dervish set up (Historical) -- deployment zones .....          | 51 |
| §9.231 -- Thorn hedge hexsides .....                                     | 52 |
| §9.232 -- Trench hexsides .....                                          | 52 |
| §9.233 -- Zariba entry/exit costs .....                                  | 52 |
| §9.321 -- British set up (Bonus) .....                                   | 53 |
| §9.322 -- Dervish enters turn one (Bonus) .....                          | 53 |
| §9.341 -- Turn 1 is always a night turn (Bonus) .....                    | 53 |
| §9.343 -- Both players use the Dervish Range Effects Table (Bonus) ..... | 53 |
| §9.344 -- Dervish controls the North Fort (Bonus) .....                  | 54 |
| §9.345 -- Gunboat White Nile <-> Blue Nile crossing (Bonus) .....        | 54 |
| §9.346 -- GORDON immobile, eliminated only at the Palace (Bonus) .....   | 54 |
| §10 -- Optional Rules .....                                              | 56 |
| §10 -- Optional Rules .....                                              | 56 |
| §10.1 -- River Mines .....                                               | 56 |
| §10.2 -- River Chain .....                                               | 56 |
| §10.11 -- Secretly record mine hexes .....                               | 57 |
| §10.12 -- Mine resolution .....                                          | 57 |
| §10.13 -- Mines consumed after both rolled for .....                     | 57 |
| §10.14 -- Dervish gunboats pass safely .....                             | 58 |
| §10.21 -- Secretly record chain hexes .....                              | 58 |
| §10.22 -- Gunboat stops on chained hex .....                             | 58 |
| §10.23 -- Sinking the chain .....                                        | 58 |
| §11 -- Historical Notes .....                                            | 59 |
| §11 -- Historical Notes .....                                            | 59 |
| Credits .....                                                            | 60 |
| §Credits -- Credits .....                                                | 60 |
| Combat Results Table (shared reference) .....                            | 61 |
| §CRT -- Combat Results Table (shared by §6.22 fire and §7.7 melee) ..... | 61 |
| Reference -- Charts and Tables .....                                     | 62 |
| §Reference -- Charts and Tables .....                                    | 62 |

|                              |    |
|------------------------------|----|
| Appendix: Symbol Index ..... | 63 |
|------------------------------|----|

0/2 implemented

## §1 -- Introduction

### §1.1 -- General Comments

descriptive

manual page 1

#### General Comments

“REMEMBER GORDON!” — THE BATTLE OF OMDURMAN is a simulation of the final battle in Great Britain’s two-year campaign to reassert her presence in the Sudan (1896–1898). Fought September 2nd, 1898, Omdurman finally broke the back of the fanatical Dervish rebellion and gained Britain a million square miles of desolate territory and two million impoverished subjects. With two players, one assumes the role of Herbert Kitchener, Sirdar (CIC) of the Anglo-Egyptian army; the other player becomes the Khalifa, Abdullah the Taiasha, absolute ruler of the Dervishes. The game is also suited for multi-player participation, with each player assuming command of one or more Dervish tribes or Anglo-Egyptian brigades.

While “REMEMBER GORDON!” — THE BATTLE OF OMDURMAN is not, strictly speaking, a beginner’s game, the mechanics of play should be familiar to players of modest experience. It is suggested that the bonus game, FALL OF KHARTOUM, and the historical scenario be played first to familiarize players with the game system prior (see manual for full text)

### §1.2 -- Game Scale

descriptive

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#### Game Scale

Each hexagon of the mapsheet represents approximately 400–440 yards of real terrain and each day turn is the equivalent of two hours of real time. Each counter of infantry and cavalry represents between 400 and 700 men, and each of the gunboats present at the battle has its own counter. The upper echelon of command is represented by individual leader counters for the Anglo-Egyptian force; and leaders plus their retinues for the Dervish army.

 3/6 implemented

## §2 -- Game Components

### §2.1 -- The Game Maps

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#### The Game Maps

The Omdurman battle map represents approximately 100 square miles of real territory and includes the area north of Omdurman in which the historical battle took place as well as the dominant terrain features that influenced the course of the battle. Note that the mapsheet also contains the Turn Record Track, Turn Sequence, and Terrain Effects Chart at the top; and the Combat Tables and Howitzer Fire Scattergram in the lower right corner. The large letters “A”, “D”, “Y”, etc. are set-up hexes for the historical scenario only (9.2) and should be ignored in the campaign game. Similarly, the hexsides of the Zariba exist only in the historical scenario and should be considered clear terrain in the campaign game. Note, however, that the Anglo-Egyptian player may “construct” the Zariba in the campaign game if desired (see 5.3). All full hexes of the Omdurman game map are playable, including the seven hexes of the Howitzer Fire Scattergram.

The mini-map for the bonus game, FALL OF KHARTOUM, represents that city as it appeared in 1885. The portion (see manual for full text)

See also: §5.3, §9.2

### §2.2 -- Play Aids

descriptive

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#### Play Aids

Certain charts and tables are needed to play the game. The Terrain Effects Chart lists all terrain found on the mapsheet and the effect of each type on movement and combat. The Combat Tables describe the range effects on various weapon types and includes the Combat Results Table. Also note the Line of Sight Table on the back of this rulebook. It tells players when certain terrain types block line of sight, thus preventing direct fire attacks on enemy units. Players should become familiar with these various charts and tables prior to the beginning of play.

(see manual for full text)

### §2.3 -- The Units

implemented

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#### The Units

| File                                                                              | Symbol              | Code Snippet                                                                                                                                          |
|-----------------------------------------------------------------------------------|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-types/src/lib.rs:654</a><br>GH:omdurman-types/src/lib.rs:654 | <b>UnitFormKind</b> | <pre>652       strum::EnumIter, 653   )] 654   pub enum UnitFormKind { 655       #[default] 656       Infantry,</pre>                                 |
| <a href="#">omdurman-rules/src/lib.rs:726</a>                                     | <b>UnitProfile</b>  | <pre>724   /// print no melee value). 725   #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 726   pub struct UnitProfile {</pre> |



|                                                                                   |                         |                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-----------------------------------------------------------------------------------|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GH:omdurman-rules/src/lib.rs:726                                                  |                         | 727   <code>pub kind: UnitKind,</code><br>728   <code>pub identity: UnitIdentity,</code>                                                                                                                                                                                                                                                                                                                         |
| <a href="#">omdurman-rules/src/lib.rs:340</a><br>GH:omdurman-rules/src/lib.rs:340 | <b>BrigadeId</b>        | 338   <code>/// (\$2.3, \$5.54). The number is the brigade ordinal as printed, e.g. `2B`.</code><br>339   <code>#[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug)]</code><br>340   <code>pub struct BrigadeId {</code><br>341   <code>pub number: u8,</code><br>342   <code>pub nationality: BrigadeNationality,</code>                                                                  |
| <a href="#">omdurman-types/src/lib.rs:806</a><br>GH:omdurman-types/src/lib.rs:806 | <b>SpriteAnnotation</b> | 804   <code>/// movement allowance (\$5.24); leaders print movement only (\$6.51).</code><br>805   <code>#[derive(Serialize, Deserialize, Clone, Debug)]</code><br>806   <code>pub struct SpriteAnnotation {</code><br>807   <code>/// Command/tribe colour. A real game indicator: Dervish leaders may only</code><br>808   <code>/// stack with units of their own colour, and different tribes may not</code> |

## §2.4 -- Game Parts Inventory

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### Game Parts Inventory

Your complete copy of “REMEMBER GORDON!” — THE BATTLE OF OMDURMAN includes:

- One 22 × 28 Battle of Omdurman mapsheet
- One 8½ × 11 bonus game: FALL OF KHARTOUM mapsheet
- One Rules Booklet
- One die-cut Unit Counter Sheet
- One Campaign Game Order of Appearance Card
- One ten-sided die
- One game box

## §2.31 -- Dervish weapon types

implemented

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Dervish artillery, gunboats, and forts fire on the “artillery” line of the Dervish Range Effects Table; Jehadia and Danagla units fire on the “rifles” line as does the Isa Zachneih unit. All other Dervish units (including leaders) are armed with spears and swords.

| File                                                                              | Symbol             | Code Snippet                                                                                                                                                                                                                                                                                                                                                            |
|-----------------------------------------------------------------------------------|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/lib.rs:502</a><br>GH:omdurman-rules/src/lib.rs:502 | <b>WeaponClass</b> | 500   <code>/// enum so a "spear" unit cannot accidentally fire on the "Howitzer" line.</code><br>501   <code>#[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug, strum::Display)]</code><br>502   <code>pub enum WeaponClass {</code><br>503   <code>/// Dervish spears and swords -- no ranged fire at all.</code><br>504   <code>Melee,</code> |
| <a href="#">omdurman-rules/src/lib.rs:946</a><br>GH:omdurman-rules/src/lib.rs:946 | <b>Howitzer</b>    | 944   <code>/// roll on the Howitzer Fire Scattergram (\$6.64).</code><br>945   <code>#[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)]</code><br>946   <code>pub struct HowitzerResolution {</code><br>947   <code>pub combat_results_table_roll: DieRoll,</code><br>948   <code>pub impact_roll: DieRoll,</code>                                   |

## §2.32 -- Anglo-Egyptian weapon types

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All Anglo-Egyptian units (except gunboats, Maxims, artillery, and leaders) are armed with rifles. Maxims fire on the “Maxims” line of the Anglo-Egyptian Range Effects Table, and artillery and old gunboats fire on the “Artillery” line. New type (named) gunboats may fire on the “Howitzer”, “Artillery”, and “Maxims” lines of the Range Effects Table. (See 6.52 for the fire capabilities of the “Friendlies”).

**\*\*Sample Dervish Units\*\*** (printed on counters): combat unit (Combat / Melee / Movement values, plus Tribe identifier); Leader (e.g. OSMAN DIGNA); Camel unit (e.g. Danagla, 4-6-12); Fort.

**\*\*Sample Anglo-Egyptian Units\*\*** (printed on counters): Cavalry (e.g. 21 Lancers); Artillery (e.g. 32 Battery); Old Gunboat (e.g. LORD KITCHENER, 0-0-15); New Gunboat — named (e.g. Sultan, with artillery and howitzer factor, plus movement downstream / movement upstream values); Maxim Guns (fire twice per turn); Infantry (Fire Combat Factor / Melee / Movement, plus Battalion ID and Brigade ID — e.g. “2B” = 2nd British Brigade, “3E” = 3rd Egyptian Brigade).

(see manual for full text)

See also: §6.52

| File                                                                              | Symbol                     | Code Snippet                                                                                                                                                                                                                                                                                                                          |
|-----------------------------------------------------------------------------------|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/lib.rs:419</a><br>GH:omdurman-rules/src/lib.rs:419 | GunboatId                  | <pre>417   /// fire; "old" gunboats do not (rulebook §2.32). 418   #[derive(Serialize, Deserialize, Clone, Copy, 419           PartialEq, Eq, Hash, Debug, strum::Display)] 420   pub enum GunboatId { 421       /// One of the five new-type named gunboats with 422       howitzer capability. 423       Named(NamedGunboat),</pre> |
| <a href="#">omdurman-rules/src/lib.rs:430</a><br>GH:omdurman-rules/src/lib.rs:430 | NamedGunboat               | <pre>428   /// The five named gunboats with howitzer capability 429   (rulebook §6.64, §2.32). 430   #[derive(Serialize, Deserialize, Clone, Copy, 431           PartialEq, Eq, Hash, Debug, strum::Display)] 432   pub enum NamedGunboat { 433       Sultan, 434       Melik,</pre>                                                  |
| <a href="#">omdurman-rules/src/lib.rs:443</a><br>GH:omdurman-rules/src/lib.rs:443 | OldGunboat                 | <pre>441   /// in the Maxim Second Fire and Howitzer subphase 442   (§6.42). 443   #[derive(Serialize, Deserialize, Clone, Copy, 444           PartialEq, Eq, Hash, Debug, strum::Display)] 445   pub enum OldGunboat { 446       LordKitchener, 447       Tamai,</pre>                                                               |
| <a href="#">omdurman-rules/src/lib.rs:443</a><br>GH:omdurman-rules/src/lib.rs:443 | GunboatId::Old             | <pre>441   /// in the Maxim Second Fire and Howitzer subphase 442   (§6.42). 443   #[derive(Serialize, Deserialize, Clone, Copy, 444           PartialEq, Eq, Hash, Debug, strum::Display)] 445   pub enum OldGunboat { 446       LordKitchener, 447       Tamai,</pre>                                                               |
| <a href="#">omdurman-rules/src/lib.rs:425</a><br>GH:omdurman-rules/src/lib.rs:425 | GunboatId::Dervish-Gunboat | <pre>423       Old(OldGunboat), 424       /// A Dervish gunboat (§9.111, §10.14). 425       DervishGunboat(u8), 426   } 427  </pre>                                                                                                                                                                                                   |

0/1 implemented

## §3 -- Getting Started

### §3 -- Getting Started

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#### Getting Started

Spread out the mapsheet on a table. It should lie flat if you backfold it against the scored lines. The Dervish player should sit next to the west edge of the map and the Anglo-Egyptian player opposite him on the east edge. Read through the rules once, looking over the various charts as they are referred to in the various sections. Next, select a scenario and punch out only those unit counters needed to play. Later on, the rest of the unit counters should be punched out, sorted and stored by unit type.

(see manual for full text)

■ 1/1 implemented

## §4 -- Turn Sequence

### §4 -- Turn Sequence

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#### Turn Sequence

“REMEMBER GORDON!” — THE BATTLE OF OMDURMAN is played in “Game Turns”, each of which has two “Player Turns”. The player moving first will vary according to the scenario being played. In the Campaign Game, for example, the Anglo-Egyptian player moves first.

### A) Anglo-Egyptian Player Turn:

1. Anglo-Egyptian Movement Phase
2. Fire Combat Phase a. Dervish Defensive Fire b. Anglo-Egyptian Offensive Fire
  1. Direct Fire Subphase
  2. Maxim Second Fire and Howitzer Fire Subphase

(see manual for full text)

| File                                                                                      | Symbol                | Code Snippet                                                                                                                                                                                                                                                                                                                                                      |
|-------------------------------------------------------------------------------------------|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/lib.rs:231</a><br>GH:omdurman-rules/src/lib.rs:231         | <b>GameTurnIndex</b>  | <pre>229   /// One-based Game Turn index (1, 2, ... up to the 230   #[derive(Serialize, Deserialize, Clone, Copy, 231   PartialEq, Eq, PartialOrd, Ord, Hash, Debug)] 232   pub struct GameTurnIndex(pub u8); 233   impl GameTurnIndex {</pre>                                                                                                                    |
| <a href="#">omdurman-rules/src/lib.rs:286</a><br>GH:omdurman-rules/src/lib.rs:286         | <b>Phase</b>          | <pre>284   /// etc. 285   #[derive(Serialize, Deserialize, Clone, Copy, 286   PartialEq, Eq, Debug, Default)] 287   pub enum Phase { 288       /// Pre-game deployment (§9.2/§9.3/§10): fixed 289       /// units are placed, each side 290       /// deploys its order of battle within its legal 291       /// zone, and river</pre>                            |
| <a href="#">omdurman-rules/src/effects.rs:385</a><br>GH:omdurman-rules/src/effects.rs:385 | <b>GameState</b>      | <pre>383   /// All mutable state of a game in progress (rulebook 384   §4). 385   #[derive(Serialize, Deserialize, Clone, Debug)] 386   pub struct GameState { 387       pub scenario: Scenario, 388       pub current_turn: GameTurnIndex,</pre>                                                                                                                 |
| <a href="#">omdurman-rules/src/effects.rs:474</a><br>GH:omdurman-rules/src/effects.rs:474 | <b>GameState::new</b> | <pre>472   impl GameState { 473       /// Create a fresh game state for a given scenario 474       /// (rulebook §4). 475       pub fn new(scenario: Scenario) -&gt; Self { 476           let first = scenario_turn(scenario, 477   GameTurnIndex(1)); 478           // First player to *move* per scenario: 479           Campaign -- Anglo-Egyptian moves</pre> |
| <a href="#">omdurman-rules/src/effects.rs:38</a><br>GH:omdurman-rules/src/effects.rs:38   | <b>AdvancePhase</b>   | <pre>36   // -- Turn / phase flow 37   // ----- 38   /// Advance to the next phase (or next player-turn 39   if melee is done) (rulebook §4). 40   AdvancePhase, 41   // -- Movement 42   -----</pre>                                                                                                                                                             |
| <a href="#">omdurman-rules/src/effects.rs:1503</a>                                        | <b>advance_phase</b>  | <pre>1501   1502   /// Advance the game state to the next phase (rulebook</pre>                                                                                                                                                                                                                                                                                   |

|                                                                             |                      |                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-----------------------------------------------------------------------------|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GH:omdurman-rules/src/effects.rs:1503                                       |                      | <pre> §4). 1503   pub fn advance_phase(state: &amp;mut GameState) -&gt; Result&lt;(), RuleError&gt; { 1504       match state.phase { 1505           // Leaving deployment is gated: both sides' required order of battle must </pre>                                                                                                                                                                                     |
| omdurman-rules/src/effects.rs:1572<br>GH:omdurman-rules/src/effects.rs:1572 | end_player_turn      | <pre> 1570   1571   /// End the current player's turn: recover disrupted units, switch active player, advance turn index (rulebook §4). 1572   pub fn end_player_turn(state: &amp;mut GameState) -&gt; Result&lt;(), RuleError&gt; { 1573       // §6.53: resolve all pending Royal Engineers demolitions before recovering 1574       // disrupted units. Each demolition checks adjacency + undisrupted status. </pre> |
| omdurman-rules/src/lib.rs:234<br>GH:omdurman-rules/src/lib.rs:234           | GameTurnIndex::value | <pre> 232   233   impl GameTurnIndex { 234       pub fn value(self) -&gt; u8 { 235           self.0 236       } </pre>                                                                                                                                                                                                                                                                                                   |
| omdurman-rules/src/effects.rs:466<br>GH:omdurman-rules/src/effects.rs:466   | PendingMelee         | <pre> 464   /// resolution after the reaction window is deterministic and host-ordered (rulebook §7.5). 465   #[derive(Serialize, Deserialize, Clone, Debug)] 466   pub struct PendingMelee { 467       pub attack: MeleeAttack, 468       pub attacker_roll: DieRoll, </pre>                                                                                                                                            |
| omdurman-rules/src/effects.rs:1322<br>GH:omdurman-rules/src/effects.rs:1322 | hex_in_enemy_zoc     | <pre> 1320       /// does not extend into or out of a Nile hex. With no board loaded these 1321       /// reduce to the plain adjacency rule. 1322       pub fn hex_in_enemy_zoc( 1323           &amp;self, 1324           hex: HexCoord, </pre>                                                                                                                                                                         |
| omdurman-rules/src/effects.rs:754<br>GH:omdurman-rules/src/effects.rs:754   | can_move_unit        | <pre> 752       /// the same `RuleError` the `MoveUnit` effect would on rejection. Lets the 753       /// UI gate input without mutating or duplicating the rules. 754       pub fn can_move_unit(&amp;self, unit_id: UnitId, cost: MovementPoints) -&gt; Result&lt;(), RuleError&gt; { 755           self.can_move_unit_to(unit_id, None, cost) 756       } </pre>                                                      |

Covered by tests: `new_game_starts_in_setup` `scenario_turn_dispatches_correctly` `turn_advances_through_phases`

 17/19 implemented

## §5 -- Movement Phase

### §5 -- Movement Phase (general)

descriptive

manual page unknown

#### Movement Phase

Covered by tests: [disrupted\\_unit\\_may\\_not\\_act](#)

### §5.3 -- Constructing the Zariba

implemented

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#### Constructing the Zariba

The Zariba trench and thorn hedge hexsides are built and in place in the historical scenario only. These hexsides are considered clear terrain in the campaign game. The Anglo-Egyptian player may, however, find it useful to construct this defensive position during the campaign game. The Zariba hexsides may only be built in their position as displayed on the mapsheet. Construction procedure is as follows: any Anglo-Egyptian infantry unit that begins and ends the Anglo-Egyptian player turn adjacent to (and on the Nile side of) Zariba hexsides has constructed all Zariba hexsides to which he is adjacent. The constructing unit may neither fire offensively nor melee attack during the turn of construction. Use a blank counter to denote units constructing Zariba hexsides. See 9.23 for defensive benefits and movement restrictions of Zariba hexsides.

(see manual for full text)

See also: §9.23

| File                                                                                        | Symbol                                     | Code Snippet                                                                                                                                                                                                                                                                                                                                                                |
|---------------------------------------------------------------------------------------------|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/lib.rs:1582</a><br>GH:omdurman-rules/src/lib.rs:1582         | <a href="#">constructing_zariba</a>        | <pre>1580             // §5.3: a unit constructing Zariba "may 1581             // neither fire offensively nor 1582             // melee attack during the turn of 1583             // construction." 1584             let s = UnitState { 1585                 constructing_zariba: true, 1586                 ..UnitState::default() 1587             }</pre>            |
| <a href="#">omdurman-rules/src/effects.rs:107</a><br>GH:omdurman-rules/src/effects.rs:107   | <a href="#">ConstructZariba</a>            | <pre>105             /// Begin constructing a Zariba hexside (rulebook 106             /// §5.3). 107             ConstructZariba { 108                 unit_ids: Vec&lt;UnitId&gt;, 109                 hexside: HexsideRef,</pre>                                                                                                                                         |
| <a href="#">omdurman-rules/src/effects.rs:2510</a><br>GH:omdurman-rules/src/effects.rs:2510 | <a href="#">apply_construct_zariba</a>     | <pre>2508             /// Mark a set of units as constructing a Zariba 2509             /// hexside (rulebook §5.3). 2510             pub fn apply_construct_zariba( 2511                 state: &amp;mut GameState, 2512                 unit_ids: &amp;[UnitId],</pre>                                                                                                    |
| <a href="#">omdurman-rules/src/lib.rs:779</a><br>GH:omdurman-rules/src/lib.rs:779           | <a href="#">UnitState::may_attack_this</a> | <pre>777             /// A unit that began construction this turn may 778             /// not fire offensively or 779             /// melee (§5.3, §6.53). 780             pub fn may_attack_this_turn(self) -&gt; bool { 781                 !self.disrupted &amp;&amp; !self.constructing_zariba 782                 &amp;&amp; !self.demolishing 783             }</pre> |

## §5.11 -- Movement allowances printed on units

implemented

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The movement allowances of the various unit types are printed directly on the units (see 2.3). A unit may move up to this printed movement allowance, paying varying costs for different terrain types (see the Terrain Effects Chart).

See also: §2.3

| File                                                                                                | Symbol                         | Code Snippet                                                                                                                                                                                                                                                                                                                                      |
|-----------------------------------------------------------------------------------------------------|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/lib.rs:112</a><br>GH:omdurman-rules/src/lib.rs:112                   | <b>MovementAllowance</b>       | <pre>110       /// is a named variant. 111       #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, PartialOrd, Ord, Hash, Debug)] 112       pub enum MovementAllowance { 113           /// Immobile (forts, wrecked gunboats). 114           Immobile = 0,</pre>                                                                       |
| <a href="#">omdurman-rules/src/lib.rs:737</a><br>GH:omdurman-rules/src/lib.rs:737                   | <b>UnitMovement</b>            | <pre>735       /// Movement allowance -- uniform for land units, split for gunboats (rulebook §5.11, §5.24, §5.25). 736       #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 737       pub enum UnitMovement { 738           Land(MovementAllowance), 739           Gunboat(GunboatMovement),</pre>                         |
| <a href="#">omdurman-types/src/lib.rs:344</a><br>GH:omdurman-types/src/lib.rs:344                   | <b>NileFlow</b>                | <pre>342       /// opposite way is <b>**upstream**</b>. 343       #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug, Default)] 344       pub struct NileFlow { 345           /// Direction the current flows toward (downstream). 346           pub dir: HexDirection,</pre>                                                     |
| <a href="#">omdurman-types/src/lib.rs:298</a><br>GH:omdurman-types/src/lib.rs:298                   | <b>HexDirection</b>            | <pre>296       /// (`+q`, `+q+r`, `+r`, `-q`, `-q-r`, `-r` for pointy- top hexes) (rulebook §5.11, §5.24). 297       #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug, Default)] 298       pub enum HexDirection { 299           #[default] 300           East = 0,</pre>                                                 |
| <a href="#">omdurman-rules/src/lib.rs:152</a><br>GH:omdurman-rules/src/lib.rs:152                   | <b>MovementPoints</b>          | <pre>150       /// Movement points spent or remaining within a single phase (rulebook §5). 151       #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, PartialOrd, Ord, Hash, Debug)] 152       pub struct MovementPoints(pub i16); 153   154       /// A distance measured in hexes (range to target, length of a retreat, ...)</pre> |
| <a href="#">omdurman-rules/src/terrain_chart.rs:21</a><br>GH:omdurman-rules/src/terrain_chart.rs:21 | <b>terrain_effects_chart</b>   | <pre>19       /// 20       /// Source: printed Terrain Effects Chart on the mapsheet. 21       pub fn terrain_effects_chart(terrain: Terrain) -&gt; TerrainEntry { 22           match terrain { 23               Terrain::Clear =&gt; TerrainEntry {</pre>                                                                                        |
| <a href="#">omdurman-rules/src/terrain_chart.rs:65</a><br>GH:omdurman-rules/src/terrain_chart.rs:65 | <b>movement_cost</b>           | <pre>63       /// Convenience: get the movement cost for a terrain type (rulebook §5.11, Terrain Effects Chart). 64       /// Returns `None` for impassable terrain (Nile). 65       pub fn movement_cost(terrain: Terrain) -&gt; Option&lt;MovementAllowance&gt; { 66           terrain_effects_chart(terrain).movement_cost 67       }</pre>    |
| <a href="#">omdurman-rules/src/terrain_chart.rs:73</a><br>GH:omdurman-                              | <b>movement_cost_with_road</b> | <pre>71       /// underlying terrain; without a road it's the terrain's own cost. The road is 72       /// a movement overlay only -- combat/LOS still use the</pre>                                                                                                                                                                              |

|                                                                                     |                                |                                                                                                                                                                                                                 |
|-------------------------------------------------------------------------------------|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| rules/src/<br>terrain_chart.rs:73                                                   |                                | <pre> underlying terrain. 73   pub fn movement_cost_with_road(terrain: Terrain, road: bool) -&gt; Option&lt;MovementAllowance&gt; { 74       if road { 75           Some(MovementAllowance::One) </pre>         |
| omdurman-<br>types/src/<br>lib.rs:430<br>GH:omdurman-<br>types/src/lib.rs:430       | Ter-<br>rain::passable_by_land | <pre> 428       } 429       /// Whether this terrain may be entered by land units (rulebook §5.11). 430       pub fn passable_by_land(self) -&gt; bool { 431           !self.is_nile() 432       } </pre>       |
| omdurman-<br>types/src/<br>lib.rs:569<br>GH:omdurman-<br>types/src/lib.rs:569       | HexData::is_crossroad          | <pre> 567           name, 568           nile_flow, 569           is_crossroad: false, 570           setup_letter: None, 571       } </pre>                                                                      |
| omdurman-<br>types/src/<br>lib.rs:569<br>GH:omdurman-<br>types/src/lib.rs:569       | TileInfo::is_crossroad         | <pre> 567           name, 568           nile_flow, 569           is_crossroad: false, 570           setup_letter: None, 571       } </pre>                                                                      |
| omdurman-<br>hexmap/src/<br>map.rs:151<br>GH:omdurman-<br>hexmap/src/<br>map.rs:151 | GameMap::roads                 | <pre> 149           tiles, 150           hexsides, 151           roads, 152           excluded: game_map.excluded.iter().map( c  (c.q, c.r)).collect(), 153           overlay: game_map.overlay.clone(), </pre> |

Covered by tests: `clear_terrain_no_bonus` `nile_is_impassable` `rough_movement_and_defense` `swamp_movement_and_defense` `hilltop_movement_and_defense` `huts_movement_and_defense` `movement_cost_convenience_matches_chart` `movement_cost_with_road_always_one` `land_unit_may_not_enter_nile` `movement_cost_without_road_matches_terrain`

## §5.12 -- Move up to allowance, hex by hex (cumulative MP cap)

implemented

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A player may move as many or as few of his units as desired during each movement phase, limited only by the units' movement allowance, the terrain costs paid in moving from hex to hex, and enemy zones of control (see 5.4).

| File                                                                                        | Symbol   | Code Snippet                                                                                                                                                                                                                                 |
|---------------------------------------------------------------------------------------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| omdurman-<br>rules/src/<br>effects.rs:1147<br>GH:omdurman-<br>rules/src/<br>effects.rs:1147 | mp_spent | <pre> 1145   1146       /// Movement points `unit_id` has already spent this turn (§5.11/§5.12). 1147       pub fn mp_spent(&amp;self, unit_id: UnitId) -&gt; i16 { 1148           self.mp_spent_this_turn 1149               .iter() </pre> |

## §5.13 -- No MP accumulation between turns

implemented

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A unit may never accumulate movement points from turn to turn, nor may a unit transfer unused movement points to other units. A unit's unused movement points in any given turn are considered lost.

| File | Symbol | Code Snippet |
|------|--------|--------------|
|------|--------|--------------|



|                                                                             |                 |                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-----------------------------------------------------------------------------|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| omdurman-rules/src/effects.rs:1572<br>GH:omdurman-rules/src/effects.rs:1572 | end_player_turn | <pre> 1570   1571   /// End the current player's turn: recover disrupted units, switch active player, advance turn index (rulebook \$4). 1572   pub fn end_player_turn(state: &amp;mut GameState) -&gt; Result&lt;(), RuleError&gt; { 1573       // \$6.53: resolve all pending Royal Engineers demolitions before recovering 1574       // disrupted units. Each demolition checks adjacency + undisrupted status.</pre> |
|-----------------------------------------------------------------------------|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## §5.21 -- Friendlies transport via gunboat

implemented

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In general, naval transport missions are not allowed, i.e. gunboats may not carry any land units. The sole exception is that the Anglo-Egyptian player may transport the surviving units of the “Friendlies” brigade from the east bank of the Nile to the west bank after, and only after, the Dervish east bank unit (Isa Zachneih) has been eliminated. The transport is accomplished in the following sequence: a) on any turn that a “Friendlies” unit and any Anglo-Egyptian gunboat start their turn adjacent, that unit may load onto (i.e. stack with) the gunboat; b) during the Anglo-Egyptian player’s next turn the gunboat may move to any Nile hex adjacent to a west bank hex (up to the gunboat’s movement allowance); c) on the Anglo-Egyptian player’s third turn the “Friendlies” unit may disembark and move normally, paying the normal terrain cost for the first hex entered. The gunboat may also move normally that turn.

(see manual for full text)

| File                                                                        | Symbol                     | Code Snippet                                                                                                                                                                                                                                                                                                                                   |
|-----------------------------------------------------------------------------|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| omdurman-rules/src/lib.rs:636<br>GH:omdurman-rules/src/lib.rs:636           | is_friendlyes              | <pre> 634       /// "Friendlies" units obey several special rules (\$5.21, \$5.23, \$6.52, 635       /// \$9.14 victory conditions). 636       pub fn is_friendlyes(&amp;self) -&gt; bool { 637           matches!( 638               self,</pre>                                                                                              |
| omdurman-rules/src/lib.rs:757<br>GH:omdurman-rules/src/lib.rs:757           | loaded_on                  | <pre> 755           pub disrupted: bool, 756           /// `Some(gunboat)` after a "Friendlies" unit loads onto a gunboat (\$5.21). 757           pub loaded_on: Option&lt;UnitId&gt;, 758           /// Set while the unit is building Zariba hexsides -- neither offensive 759           /// fire nor melee allowed that turn (\$5.3).</pre> |
| omdurman-rules/src/effects.rs:134<br>GH:omdurman-rules/src/effects.rs:134   | FriendliesTransport        | <pre> 132   133       /// Load/disembark the "Friendlies" brigade via gunboat (rulebook \$5.21). 134       FriendliesTransport(crate::FriendliesTransport), 135   136       // -- Optional rules -----</pre>                                                                                                                                   |
| omdurman-rules/src/effects.rs:2769<br>GH:omdurman-rules/src/effects.rs:2769 | apply_friendlyes_transport | <pre> 2767       /// transport mission ends and the unit is freed (a disembarking `MoveUnit` 2768       /// should follow, costed by terrain). 2769       pub fn apply_friendlyes_transport( 2770           state: &amp;mut GameState, 2771           action: FriendliesTransport,</pre>                                                       |
| omdurman-rules/src/lib.rs:1023<br>GH:omdurman-rules/src/lib.rs:1023         | FriendliesTransport        | <pre> 1021       /// tracks each unit-gunboat pair independently. 1022       #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 1023       pub enum FriendliesTransport { 1024           /// Turn N (the load turn): unit and gunboat</pre>                                                                                  |

|  |  |                                                                                          |
|--|--|------------------------------------------------------------------------------------------|
|  |  | <pre> started adjacent; unit 1025       /// loads onto (stacks with) the gunboat. </pre> |
|--|--|------------------------------------------------------------------------------------------|

## §5.22 -- Land units may never enter a Nile River hex

implemented

manual page unknown

With the exception of 5.21, land units may never enter a Nile River hex. Only gunboats may enter and move along Nile River hexes.

See also: §5.21

| File                                                                                      | Symbol                  | Code Snippet                                                                                                                                                                                             |
|-------------------------------------------------------------------------------------------|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/effects.rs:771</a><br>GH:omdurman-rules/src/effects.rs:771 | <b>can_move_unit_to</b> | <pre> 769       /// 770       /// [<code>hex_in_enemy_zoc</code>]: <code>Self::hex_in_enemy_zoc</code> 771       pub fn can_move_unit_to( 772           &amp;self, 773           unit_id: UnitId, </pre> |

## §5.23 -- Walled city entry restrictions

implemented

manual page unknown

Only certain units may enter the walled portion of Omdurman. For the Dervish player these are the Khalifa unit, the three Dervish artillery units, and the Taiasha units (the Khalifa's bodyguard). Any Anglo-Egyptian units that can get to the walled city may enter it (except gunboats and "Friendlies"). Units entering and/or exiting the walled city may only do so through a gate or breach hexside.

| File                                                                                        | Symbol                          | Code Snippet                                                                                                                                                                                                                                                                                                                                                                                                  |
|---------------------------------------------------------------------------------------------|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-types/src/lib.rs:158</a><br>GH:omdurman-types/src/lib.rs:158           | <b>HexsideRef</b>               | <pre> 156       /// data by [<code>HexsideRef</code>]. 157       #[derive(Serialize, Deserialize, Clone, Copy, 158           PartialEq, Eq, Hash, Debug)] 159       pub struct HexsideRef { 160           pub a: HexCoord, </pre>                                                                                                                                                                             |
| <a href="#">omdurman-types/src/lib.rs:193</a><br>GH:omdurman-types/src/lib.rs:193           | <b>HexsideKind</b>              | <pre> 191           strum::EnumIter, 192       )] 193       pub enum HexsideKind { 194           /// City wall (Khartoum, walled city of Omdurman). 195           Blocks LOS, blocks 196           /// movement except across gates/breaches (§5.23), 197           blocks ZOC into the city </pre>                                                                                                           |
| <a href="#">omdurman-types/src/lib.rs:257</a><br>GH:omdurman-types/src/lib.rs:257           | <b>blocks_movement</b>          | <pre> 255       /// Whether land movement may *not* cross this side 256       (§5.23). Walls block 257       /// movement except at gates/breaches. 258       pub fn blocks_movement(self) -&gt; bool { 259           matches!(self, HexsideKind::Wall) </pre>                                                                                                                                                |
| <a href="#">omdurman-rules/src/effects.rs:2329</a><br>GH:omdurman-rules/src/effects.rs:2329 | <b>can_retreat_before_melee</b> | <pre> 2327       /// two hexes away and empty. (Does not verify the 2328       attacker is infantry -- 2329       /// the caller offers the retreat only in response 2330       to one.) 2331       pub fn can_retreat_before_melee(&amp;self, unit_id: 2332           UnitId, to: HexCoord) -&gt; Result&lt;(), RuleError&gt; { 2333           let unit = self 2334               .find_unit(unit_id) </pre> |

## §5.24 -- Gunboat upstream/downstream movement

implemented

manual page unknown

Note that gunboats have two movement allowances separated by a slash, e.g. 10/16. The smaller number is the movement allowance when moving upstream, i.e. against the current (the direction of the current is indicated by arrows in the Nile). The larger number is the movement allowance when moving downstream, i.e. with the current. Gunboats may combine movement in both directions, but if they move even one hex upstream, their upstream movement allowance is their maximum movement allowance for that turn, and may not be exceeded.

(see manual for full text)

| File                                                                              | Symbol                 | Code Snippet                                                                                                                                                                                                                         |
|-----------------------------------------------------------------------------------|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-types/src/lib.rs:344</a><br>GH:omdurman-types/src/lib.rs:344 | <b>NileFlow</b>        | <pre>342   /// opposite way is **upstream**.<br/>343   #[derive(Serialize, Deserialize, Clone, Copy,<br/>344   pub struct NileFlow {<br/>345       /// Direction the current flows toward<br/>346       pub dir: HexDirection,</pre> |
| <a href="#">omdurman-rules/src/lib.rs:570</a><br>GH:omdurman-rules/src/lib.rs:570 | <b>GunboatMovement</b> | <pre>568   /// the turn.<br/>569   #[derive(Serialize, Deserialize, Clone, Copy,<br/>570   pub struct GunboatMovement {<br/>571       pub upstream: MovementAllowance,<br/>572       pub downstream: MovementAllowance,</pre>        |
| <a href="#">omdurman-types/src/lib.rs:672</a><br>GH:omdurman-types/src/lib.rs:672 | <b>is_boat</b>         | <pre>670   impl UnitFormKind {<br/>671       /// Gunboats use the split upstream/downstream<br/>672       pub fn is_boat(self) -&gt; bool {<br/>673           matches!(self, UnitFormKind::Gunboat)<br/>674       }</pre>            |

## §5.25 -- Dervish forts may not move

implemented

manual page unknown

Dervish forts may not move in any way once placed.

| File                                                                              | Symbol          | Code Snippet                                                                                                                                                                                                      |
|-----------------------------------------------------------------------------------|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/lib.rs:114</a><br>GH:omdurman-rules/src/lib.rs:114 | <b>Immobile</b> | <pre>112       pub enum MovementAllowance {<br/>113           /// Immobile (forts, wrecked gunboats).<br/>114           Immobile = 0,<br/>115           One = 1,<br/>116           Two = 2,</pre>                 |
| <a href="#">omdurman-rules/src/lib.rs:469</a><br>GH:omdurman-rules/src/lib.rs:469 | <b>Fort</b>     | <pre>467       Gunboat,<br/>468       /// Permanent emplacement -- may not move once<br/>469       Fort,<br/>470       /// Dervish leader: has fire/melee/movement factors<br/>471       DervishLeaderUnit,</pre> |

|                                                                                   |                               |                                                                                                                                      |
|-----------------------------------------------------------------------------------|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/lib.rs:741</a><br>GH:omdurman-rules/src/lib.rs:741 | <b>UnitMovement::Immobile</b> | <pre> 739   Gunboat(GunboatMovement), 740       /// Forts may not move once placed (§5.25). 741       Immobile, 742   } 743   </pre> |
|-----------------------------------------------------------------------------------|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|

## §5.26 -- Units stop on entering enemy ZOC

implemented

manual page unknown

Units must stop their movement immediately upon entering an enemy zone of control (see 5.4).

| File                                                                                      | Symbol                  | Code Snippet                                                                                                                                                                                 |
|-------------------------------------------------------------------------------------------|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/effects.rs:771</a><br>GH:omdurman-rules/src/effects.rs:771 | <b>can_move_unit_to</b> | <pre> 769       /// 770       /// [<code>hex_in_enemy_zoc`</code>]: Self::hex_in_enemy_zoc 771       pub fn can_move_unit_to( 772           &amp;self, 773           unit_id: UnitId, </pre> |

## §5.41 -- All units except AE leaders exert ZOC

implemented

manual page unknown

All units except Anglo-Egyptian leaders exert a zone of control (hereafter called a ZOC) into their six adjacent hexes (exception: Gunboats exert a ZOC only against enemy gunboats). Disrupted units have no ZOC.

| File                                                                                        | Symbol                   | Code Snippet                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|---------------------------------------------------------------------------------------------|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/lib.rs:812</a><br>GH:omdurman-rules/src/lib.rs:812           | <b>ZocReason</b>         | <pre> 810       /// Used by the engine when answering "is this hex in             an enemy ZOC?". 811       #[derive(Serialize, Deserialize, Clone, Copy,                     PartialEq, Eq, Debug)] 812       pub enum ZocReason { 813           /// Normal ZOC: any non-disrupted unit other than                 an Anglo-Egyptian 814           /// leader (§5.41) projects ZOC into each of its                 six adjacent hexes. </pre> |
| <a href="#">omdurman-rules/src/effects.rs:1287</a><br>GH:omdurman-rules/src/effects.rs:1287 | <b>unit_projects_zoc</b> | <pre> 1285       /// §5.44) need the game map, which the engine             does not hold; the app layers 1286       /// those on top. This is the position/kind/             disruption core of the rule. 1287       pub fn unit_projects_zoc( 1288           &amp;self, 1289           unit: &amp;UnitPlacement, </pre>                                                                                                                       |

## §5.42 -- No MP cost to enter/leave enemy ZOC

implicit

manual page unknown

There is no movement point cost to enter or leave an enemy ZOC.

Covered by tests: [entering\\_enemy\\_zoc\\_costs\\_no\\_extra\\_mp](#)

## §5.43 -- Units stop when entering enemy ZOC

implemented

manual page unknown

All units must stop when they enter an enemy ZOC and may move no further that turn. In their next movement phase they may withdraw or, if desired, move directly into another enemy ZOC.

| File                                                                                        | Symbol                        | Code Snippet                                                                                                                                                                                                                                  |
|---------------------------------------------------------------------------------------------|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/effects.rs:771</a><br>GH:omdurman-rules/src/effects.rs:771   | <code>can_move_unit_to</code> | <pre>769      /// 770      /// [<code>hex_in_enemy_zoc</code>]: Self::hex_in_enemy_zoc 771      pub fn can_move_unit_to( 772          &amp;self, 773          unit_id: UnitId,</pre>                                                          |
| <a href="#">omdurman-rules/src/effects.rs:1322</a><br>GH:omdurman-rules/src/effects.rs:1322 | <code>hex_in_enemy_zoc</code> | <pre>1320      /// does not extend into or out of a Nile hex.     With no board loaded these 1321      /// reduce to the plain adjacency rule. 1322      pub fn hex_in_enemy_zoc( 1323          &amp;self, 1324          hex: HexCoord,</pre> |

## §5.44 -- ZOC limitations (walls, khor, fort, Nile, Zariba)

implemented

manual page unknown

ZOCs do not extend into or out of a Nile River hex (exception: Gunboats, see 5.41). ZOCs do not extend across a khor, into a fort, or into a hex inside the walled city across a wall hexside. ZOCs do extend out of a fort (even if unoccupied), and from a walled city hex into an adjacent non-walled-city hex across a wall hexside. ZOCs also extend out of (but not into) a walled city hex across a gate hexside. ZOCs extend both ways across a breach hexside. ZOCs also extend out of, but not into, a hut or building hex. In the historical scenario ZOCs extend out of, but not into, the Zariba across a Zariba hexside (also in the campaign game if the Zariba is constructed).

(see manual for full text)

See also: §5.41

| File                                                                              | Symbol                 | Code Snippet                                                                                                                                                                                                                                                                                                                                                                             |
|-----------------------------------------------------------------------------------|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/lib.rs:812</a><br>GH:omdurman-rules/src/lib.rs:812 | <code>ZocReason</code> | <pre>810      /// Used by the engine when answering "is this hex in     an enemy ZOC?". 811      #[derive(Serialize, Deserialize, Clone, Copy,     PartialEq, Eq, Debug)] 812      pub enum ZocReason { 813          /// Normal ZOC: any non-disrupted unit other than     an Anglo-Egyptian 814          /// leader (§5.41) projects ZOC into each of its     six adjacent hexes.</pre> |
| <a href="#">omdurman-types/src/lib.rs:198</a><br>GH:omdurman-types/src/lib.rs:198 | <code>Wall</code>      | <pre>196      /// (§5.44), blocks melee (§7.2), blocks advance-     after-combat (§6.82). 197      #[default] 198      Wall, 199      /// Gate hexside in a wall. ZOC extends *out of*     the walled city through 200      /// gates but not into it (§5.44). Melee may be     made through a gate (§7.2).</pre>                                                                        |

|                                                                                             |                   |                                                                                                                                                                                                                                                                                                                                  |
|---------------------------------------------------------------------------------------------|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-types/src/lib.rs:207</a><br>GH:omdurman-types/src/lib.rs:207           | Khor              | <pre> 205       /// Khor -- gully/wadi. ZOCs do not extend across           (\$5.44); advance after 206       /// combat may not cross (\$6.82). 207       Khor, 208       /// Crest line. Blocks LOS unless the firer is on           the higher side 209       /// (\$6.3 note 7).           </pre>                            |
| <a href="#">omdurman-rules/src/lib.rs:825</a><br>GH:omdurman-rules/src/lib.rs:825           | ZocReason::Zariba | <pre> 823       /// Zariba hexside ZOC behaviour in the historical           scenario / when the 824       /// Zariba is constructed (\$5.44). 825       Zariba, 826   } 827             </pre>                                                                                                                                  |
| <a href="#">omdurman-rules/src/effects.rs:1287</a><br>GH:omdurman-rules/src/effects.rs:1287 | unit_projects_zoc | <pre> 1285       /// \$5.44) need the game map, which the engine           does not hold; the app layers 1286       /// those on top. This is the position/kind/           disruption core of the rule. 1287       pub fn unit_projects_zoc( 1288           &amp;self, 1289           unit: &amp;UnitPlacement,           </pre> |

## §5.51 -- Stacking limit (4 units + leaders, gunboats isolated)

implemented

manual page unknown

No more than four units may occupy a hex, with the exception of leaders and gunboats. All leader units are free stacking, i.e. they may stack in addition to the four-unit-per-hex stacking limitation. Gunboats may not stack with any other unit (Exception: 5.21). Players may move through friendly units at no additional cost in movement points. The stacking limitation applies only at the end of the movement phase and during combat.

See also: §5.21

| File                                                                              | Symbol       | Code Snippet                                                                                                                                                                                                                                                                                                                                                                                              |
|-----------------------------------------------------------------------------------|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/lib.rs:834</a><br>GH:omdurman-rules/src/lib.rs:834 | OverLimit    | <pre> 832       /// and the gunboat exception. 833       #[error("hex stack exceeds the four-unit limit")] 834       OverLimit, 835       /// "Gunboats may not stack with any other           unit" (\$5.51, exception §5.21). 836       #[error("gunboats may not stack with non-gunboat           units")]           </pre>                                                                            |
| <a href="#">omdurman-rules/src/lib.rs:837</a><br>GH:omdurman-rules/src/lib.rs:837 | GunboatStack | <pre> 835       /// "Gunboats may not stack with any other           unit" (\$5.51, exception §5.21). 836       #[error("gunboats may not stack with non-gunboat           units")] 837       GunboatStack, 838       /// "Units of different Dervish tribes may not           stack together" (\$5.52). 839       #[error("Dervish units of different tribes may not           stack")]           </pre> |

Covered by tests: [stacking\\_over\\_limit\\_rejected](#)

## §5.52 -- Different Dervish tribes may not stack together

implemented

manual page unknown

The units of different Dervish tribes may not stack together, even if they are the same color (e.g. although both are green, Mulazmin and Jehadia units may not stack with each other).

| File                                                                              | Symbol                        | Code Snippet                                                                                                                                                                                                                                                                                                                                             |
|-----------------------------------------------------------------------------------|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/lib.rs:840</a><br>GH:omdurman-rules/src/lib.rs:840 | <b>DervishTribeMix</b>        | <pre> 838       /// "Units of different Dervish tribes may not       stack together" (§5.52). 839       #[error("Dervish units of different tribes may not       stack")] 840       DervishTribeMix, 841       /// "If Dervish leaders elect to stack, they may       only stack with units of 842       /// their command (i.e. colour)" (§5.53).</pre> |
| <a href="#">omdurman-types/src/lib.rs:466</a><br>GH:omdurman-types/src/lib.rs:466 | <b>Terrain::overlay_color</b> | <pre> 464   465       /// Return an RGBA colour suitable for a terrain-       type overlay. 466       pub fn overlay_color(self) -&gt; [f32; 4] { 467           self.color().rgba() 468       }</pre>                                                                                                                                                    |

## §5.53 -- Leader stacking with command colour only

implemented

manual page unknown

Leader units are not required to stack. If Dervish leaders elect to stack, however, they may only stack with units of their command (i.e. color). For example, Sheik El Din may only stack with Mulazmins or Jehadias.

**\*\*5.54) Anglo-Egyptian Brigade Integrity:\*\*** All British, Sudanese, and Egyptian infantry units have their brigade designation printed in the upper right corner (e.g. “2B” = 2nd British Brigade; “3E” = 3rd Egyptian Brigade, etc.). In any combat phase in which all four infantry battalions belonging to any Anglo-Egyptian infantry brigade are stacked in the same hex they are said to have brigade integrity. Stacks having brigade integrity receive a +1 modifier to their fire combat die roll provided they all fire at the same enemy occupied hex. This modifier is in addition to the normal +1 bonus given to all Anglo-Egyptian direct fire attacks (see 6.24).

(see manual for full text)

See also: §5.54, §6.24

| File                                                                              | Symbol                              | Code Snippet                                                                                                                                                                                                    |
|-----------------------------------------------------------------------------------|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/lib.rs:844</a><br>GH:omdurman-rules/src/lib.rs:844 | <b>DervishLeaderCommandMismatch</b> | <pre> 842       /// their command (i.e. colour)" (§5.53). 843       #[error("Dervish leader may only stack with units       of their own command")] 844       DervishLeaderCommandMismatch, 845   } 846  </pre> |

## §5.54 -- Anglo-Egyptian Brigade Integrity

implemented

manual page unknown

| File                                                                              | Symbol                   | Code Snippet                                                                                                                                                                                                                                                                               |
|-----------------------------------------------------------------------------------|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/lib.rs:850</a><br>GH:omdurman-rules/src/lib.rs:850 | <b>BrigadeIntegrity</b>  | <pre> 848       /// stack contains all four battalions of a single       Anglo-Egyptian brigade. 849       #[derive(Serialize, Deserialize, Clone, Copy,       PartialEq, Eq, Debug)] 850       pub enum BrigadeIntegrity { 851           None, 852           Integrated(BrigadeId),</pre> |
| <a href="#">omdurman-rules/src/lib.rs:699</a>                                     | <b>brigade_integrity</b> | <pre> 697       /// Only a full stack of four battalions qualifies.       Three or fewer may still 698       /// stack and fire, but they receive no brigade-</pre>                                                                                                                        |

|                                                                                   |                  |                                                                                                                                                                                                                                                                                           |
|-----------------------------------------------------------------------------------|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GH:omdurman-rules/src/lib.rs:699                                                  |                  | <pre> integrity bonus. 699   pub fn brigade_integrity(identities: &amp;[UnitIdentity]) - &gt; BrigadeIntegrity { 700       let Some(brigade) = identities.first().and_then( i  i.brigade()) else { 701           return BrigadeIntegrity::None; </pre>                                    |
| <a href="#">omdurman-rules/src/lib.rs:349</a><br>GH:omdurman-rules/src/lib.rs:349 | BattalionOrdinal | <pre> 347       /// brigade integrity requires all four stacked in one hex (\$5.54). 348       #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug, strum::Display)] 349       pub enum BattalionOrdinal { 350           First = 1, 351           Second = 2, </pre> |
| <a href="#">omdurman-types/src/lib.rs:712</a><br>GH:omdurman-types/src/lib.rs:712 | Brigade          | <pre> 710       Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug, strum::Display, strum::EnumIter, 711   )] 712   pub enum BrigadeNationality { 713       /// `xB` -- British. 714       British, </pre>                                                                   |

 21/25 implemented



## §6 -- Fire Combat Phase

### §6.3 -- Line of Sight Table

implemented

manual page unknown

#### Line of Sight Table

This table is located on the back of this rulebook and should be self-explanatory. Locate the terrain type the firing unit is in and cross-index it with the terrain type the target unit is in. Terrain types in the intersecting box block line of sight, with exceptions as footnoted. Also study the “Special LOS Notes” given and remember that (with the exception of howitzer fire — see 6.64) you can’t fire at anything you can’t see!

See also: §6.64

| File                                                                                        | Symbol                     | Code Snippet                                                                                                                                                                                                                                                                                               |
|---------------------------------------------------------------------------------------------|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/los_table.rs:3</a><br>GH:omdurman-rules/src/los_table.rs:3   | <b>LosFirerTerrain</b>     | <pre>1   /// Terrain type of the *firing* unit's hex for LOS     purposes (rulebook §6.3). 2   #[derive(Clone, Copy, PartialEq, Eq, Hash, Debug)] 3   pub enum LosFirerTerrain { 4       Ground, 5       Rough,</pre>                                                                                      |
| <a href="#">omdurman-rules/src/los_table.rs:11</a><br>GH:omdurman-rules/src/los_table.rs:11 | <b>LosTargetTerrain</b>    | <pre>9   /// Terrain type of the *target* unit's hex for LOS     purposes (rulebook §6.3). 10   #[derive(Clone, Copy, PartialEq, Eq, Hash, Debug)] 11   pub enum LosTargetTerrain { 12       Ground, 13       /// Units in the hex (including friendly -- LOS is     blocked if the</pre>                  |
| <a href="#">omdurman-rules/src/los_table.rs:27</a><br>GH:omdurman-rules/src/los_table.rs:27 | <b>LosResult</b>           | <pre>25   /// Whether LOS is blocked (rulebook §6.3). 26   #[derive(Clone, Copy, PartialEq, Eq, Debug)] 27   pub enum LosResult { 28       Clear, 29       Blocked,</pre>                                                                                                                                  |
| <a href="#">omdurman-rules/src/los_table.rs:37</a><br>GH:omdurman-rules/src/los_table.rs:37 | <b>los_table</b>           | <pre>35   /// If the cell says "Blocks", LOS is blocked;     otherwise it is clear 36   /// (subject to the special notes below). 37   pub fn los_table(firer: LosFirerTerrain, target:     LosTargetTerrain) -&gt; LosResult { 38       use LosFirerTerrain as F; 39       use LosResult::*;</pre>        |
| <a href="#">omdurman-rules/src/los_table.rs:89</a><br>GH:omdurman-rules/src/los_table.rs:89 | <b>LosSpecialNote</b>      | <pre>87   /// 7. Crest hexsides block LOS unless the firer is on     the higher side 88   /// of the crest. 89   pub enum LosSpecialNote { 90       MaxTwoTreeHutHexes, 91       HilltopToHilltop,</pre>                                                                                                   |
| <a href="#">omdurman-types/src/lib.rs:234</a><br>GH:omdurman-types/src/lib.rs:234           | <b>blocks_los</b>          | <pre>232   /// Whether this hexside blocks line of sight     across it (§6.3). Crest is 233   /// directional and handled by the caller; here it     is treated as blocking. 234   pub fn blocks_los(self) -&gt; bool { 235       matches!(self, HexsideKind::Wall       HexsideKind::Crest) 236   }</pre> |
| <a href="#">omdurman-types/src/lib.rs:436</a>                                               | <b>Terrain::blocks_los</b> | <pre>434   /// Whether an intervening hex of this terrain     unconditionally blocks line 435   /// of sight (§6.3). 436   pub fn blocks_los(self) -&gt; bool {</pre>                                                                                                                                      |

|                                  |  |                                                                                         |
|----------------------------------|--|-----------------------------------------------------------------------------------------|
| GH:omdurman-types/src/lib.rs:436 |  | <pre> 437           matches!(self, Terrain::Huts   Terrain::Building) 438       }</pre> |
|----------------------------------|--|-----------------------------------------------------------------------------------------|

Covered by tests: `los_wall_blocks` `los_ground_to_ground_clear` `los_hilltop_to_huts_blocked` `ground_firer_covers_all_targets` `rough_firer_covers_all_targets` `hilltop_firer_covers_all_targets` `all_24_cells_exercised`

## §6.6 -- Special Artillery Capabilities

implemented

manual page unknown

### Special Artillery Capabilities

| File                                                                              | Symbol             | Code Snippet                                                                                                                                                                                                                                                                                                   |
|-----------------------------------------------------------------------------------|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/lib.rs:502</a><br>GH:omdurman-rules/src/lib.rs:502 | <b>WeaponClass</b> | <pre> 500   /// enum so a "spear" unit cannot accidentally fire on the "Howitzer" line. 501   #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug, strum::Display)] 502   pub enum WeaponClass { 503       /// Dervish spears and swords -- no ranged fire at all. 504       Melee,</pre> |

## §6.7 -- Defensive Fire

implemented

manual page unknown

### Defensive Fire

In Defensive Fire phase, all of the non-moving player's units may fire at any of the moving player's units in range, within the limitations imposed by the rules of combat (see 6.1 to 6.6). There is no advance after combat as a result of defensive fires.

See also: §6.6

| File                                                                                        | Symbol                          | Code Snippet                                                                                                                                                                                                                                                                                                                                                                        |
|---------------------------------------------------------------------------------------------|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/effects.rs:2374</a><br>GH:omdurman-rules/src/effects.rs:2374 | <b>can_advance_after_combat</b> | <pre> 2372       /// player's unit, not artillery, adjacent to `to`, and `to` now empty. 2373       /// Wall/khor hexside restrictions are not enforced (no hexside map data). 2374       pub fn can_advance_after_combat(&amp;self, unit_id: UnitId, to: HexCoord) -&gt; Result&lt;(), RuleError&gt; { 2375           let unit = self 2376               .find_unit(unit_id)</pre> |

## §6.11 -- Fire combat factor printed on units

implemented

manual page unknown

The fire combat factor of the various unit types is printed directly on the units and is a numerical expression of the unit's fire strength.

| File                                | Symbol            | Code Snippet                                                                                     |
|-------------------------------------|-------------------|--------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/</a> | <b>FireFactor</b> | <pre> 65       /// Every possible value from the annotated counter set is a named variant.</pre> |

|                                              |  |                                                                                                                                                                                                                               |
|----------------------------------------------|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| lib.rs:67<br>GH:omdurman-rules/src/lib.rs:67 |  | <pre> 66       #[derive(Serialize, Deserialize, Clone, Copy, 67               PartialEq, Eq, PartialOrd, Ord, Hash, Debug, strum::Display)] 68       pub enum FireFactor { 69           One = 1,            Three = 3, </pre> |
|----------------------------------------------|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## §6.12 -- Fire combat is always voluntary

implicit

manual page unknown

Fire combat is always voluntary. A unit is never required to fire at enemy units merely because they are in range or adjacent.

## §6.13 -- Fire factor is unitary (may not be divided)

implicit

manual page unknown

If a unit elects to fire, its fire combat factor at an enemy unit, that fire combat factor is unitary. A unit's fire combat factor may not be divided up to fire at enemy units on different hexes.

## §6.14 -- Players may combine fire factors into one attack

implemented

manual page unknown

Players may combine fire during fire combat phase, i.e. they may fire at an enemy-occupied hex with as many friendly units as may legally do so, combining all of their fire combat factors into one attack. Note that in any given fire combat phase, however, a combat unit may only fire once and may only be fired at once (exceptions: Maxim guns and gunboats — see 6.4).

| File                                                            | Symbol     | Code Snippet                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-----------------------------------------------------------------|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| omdurman-rules/src/lib.rs:81<br>GH:omdurman-rules/src/lib.rs:81 | sum_to_row | <pre> 79   impl FireFactor { 80       /// Sum multiple fire factors and return the       corresponding Combat Results Table row (rulebook §6.11). 81       pub fn sum_to_row&lt;'a&gt;(factors: impl       IntoIterator&lt;Item = &amp;'a FireFactor&gt;) -&gt; FireFactorRow { 82           let total: u16 = factors.into_iter().map( f        f.value()).sum(); 83           crate::combat_results_table::FireFactorRow::from_total(total) </pre> |

## §6.15 -- May divide a stack to fire at different hexes

implicit

manual page unknown

Players may also divide a stack of units in order to fire at different enemy-occupied hexes. Anglo-Egyptian infantry units having brigade integrity, however, do not receive their +1 direct fire modifier unless they all fire at the same enemy-occupied hex (see 5.54).

See also: §5.54

## §6.16 -- Halving fire strength rounds down, minimum 1

implemented

manual page unknown

When halving fire combat strength, always round down each individual unit. For example, an Egyptian brigade of four battalions, each having a printed strength of 9 fire factors, will fire a total of 16 factors when halved. However, a unit's firing strength is never reduced below one by halving.

| File                                                                              | Symbol    | Code Snippet                                                                                                                                                                                              |
|-----------------------------------------------------------------------------------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/lib.rs:537</a><br>GH:omdurman-rules/src/lib.rs:537 | RangeBand | <pre> 535   /// multiplied at a given distance (§6.22). 536   #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 537   pub enum RangeBand { 538       Tripled, 539       Doubled,</pre> |

## §6.21 -- First check LOS before firing

implemented

manual page unknown

When combat units wish to fire at enemy units, first check the Line of Sight Table to be sure the firing unit can see the target hex (exception: howitzer fire, see 6.64).

See also: §6.64

| File                                                                                        | Symbol    | Code Snippet                                                                                                                                                                                                                                                                                 |
|---------------------------------------------------------------------------------------------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/los_table.rs:37</a><br>GH:omdurman-rules/src/los_table.rs:37 | los_table | <pre> 35   /// If the cell says "Blocks", LOS is blocked; otherwise it is clear 36   /// (subject to the special notes below). 37   pub fn los_table(firer: LosFirerTerrain, target: LosTargetTerrain) -&gt; LosResult { 38       use LosFirerTerrain as F; 39       use LosResult::*;</pre> |

## §6.22 -- Consult Range Effects Table

implemented

manual page unknown

Next consult the Range Effects Table to see if the firing unit's fire combat factor is tripled, doubled, normal, halved, or if the target hex is out of range. Add up the total number of fire combat factors firing at the enemy-occupied hex.

| File                                                                                                | Symbol                | Code Snippet                                                                                                                                                                                                                                                                                                                       |
|-----------------------------------------------------------------------------------------------------|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/range_effects.rs:23</a><br>GH:omdurman-rules/src/range_effects.rs:23 | ae_range_effects      | <pre> 21   /// Look up the range band for an Anglo-Egyptian weapon (§6.22, §6.24). 22   /// Distances &gt; 10 are out of range for all weapons. 23   pub fn ae_range_effects(weapon: WeaponClass, distance: HexDistance) -&gt; RangeBand { 24       if distance.value() &gt; 10 { 25           return RangeBand::OutOfRange;</pre> |
| <a href="#">omdurman-rules/src/range_effects.rs:61</a><br>GH:omdurman-rules/src/range_effects.rs:61 | dervish_range_effects | <pre> 59   /// Look up the range band for a Dervish weapon (§6.22). 60   /// Distances &gt; 10 are out of range for all weapons. 61   pub fn dervish_range_effects(weapon: WeaponClass, distance: HexDistance) -&gt; RangeBand { 62       if distance.value() &gt; 10 { 63           return RangeBand::OutOfRange;</pre>           |
| <a href="#">omdurman-rules/src/lib.rs:521</a><br>GH:omdurman-rules/src/lib.rs:521                   | Range                 | <pre> 519   /// (rulebook §6.22). Distances beyond 10 hexes are out of range for all weapons. 520   #[derive(Clone, Copy, PartialEq, Eq, Debug)] 521   pub enum Range {</pre>                                                                                                                                                      |

|                                                                                   |             |                                                                                                                                                                                                                       |
|-----------------------------------------------------------------------------------|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                   |             | <pre> 522   One, 523   Two,</pre>                                                                                                                                                                                     |
| <a href="#">omdurman-rules/src/lib.rs:537</a><br>GH:omdurman-rules/src/lib.rs:537 | RangeBand   | <pre> 535   /// multiplied at a given distance (§6.22). 536   #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 537   pub enum RangeBand { 538       Tripled, 539       Doubled,</pre>             |
| <a href="#">omdurman-rules/src/lib.rs:157</a><br>GH:omdurman-rules/src/lib.rs:157 | HexDistance | <pre> 155   /// (rulebook §6.22, §7.5). 156   #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, PartialOrd, Ord, Hash, Debug)] 157   pub struct HexDistance(pub u16); 158   159   impl HexDistance {</pre> |

Covered by tests: [ae\\_rifles\\_doubled\\_at\\_range\\_1](#) [ae\\_rifles\\_halved\\_at\\_range\\_4](#) [ae\\_howitzer\\_range](#) [dervish\\_rifles\\_shorter\\_range](#) [melee\\_only\\_range\\_1](#) [ae\\_range\\_effects\\_artillery\\_full](#) [ae\\_range\\_effects\\_maxims\\_match\\_rifles](#) [ae\\_range\\_effects\\_distance\\_over\\_10](#) [ae\\_range\\_effects\\_howitzer\\_halved\\_4\\_to\\_10](#) [dervish\\_range\\_effects\\_rifles](#) [dervish\\_range\\_effects\\_artillery](#) [dervish\\_range\\_effects\\_maxims\\_and\\_howitzer](#) [dervish\\_range\\_effects\\_melee](#) [dervish\\_range\\_effects\\_distance\\_over\\_10](#) [fire\\_combat\\_eliminate\\_target](#) [max\\_day\\_range\\_all\\_combos](#)

## §6.23 -- Terrain defensive modifier

implemented

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Next check the Terrain Effects Chart to see if the enemy-occupied hex fired upon contains any terrain which gives the enemy units in that hex a defensive benefit. If so, apply this negative modifier to the roll of the ten-sided die and cross-index your net die roll on the Combat Results Table with the number of combat factors firing.

| File                                                                                                | Symbol                | Code Snippet                                                                                                                                                                                                                                                                                                        |
|-----------------------------------------------------------------------------------------------------|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/terrain_chart.rs:59</a><br>GH:omdurman-rules/src/terrain_chart.rs:59 | defense_modifier      | <pre> 57   58   /// Convenience: get the defense modifier for a terrain type (rulebook §6.23, Terrain Effects Chart). 59   pub fn defense_modifier(terrain: Terrain) -&gt; i16 { 60       terrain_effects_chart(terrain).defense_modifier 61   }</pre>                                                              |
| <a href="#">omdurman-rules/src/lib.rs:871</a><br>GH:omdurman-rules/src/lib.rs:871                   | Terrain               | <pre> 869       /// Negative modifier from the Terrain Effects Chart applied to the 870       /// defender's hex (§6.23). 871       Terrain(i16), 872       /// -2 thorn-hedge defensive modifier (§9.231). 873       ZaribaThornHedge,</pre>                                                                       |
| <a href="#">omdurman-types/src/lib.rs:234</a><br>GH:omdurman-types/src/lib.rs:234                   | Terrain::blocks_los   | <pre> 232       /// Whether this hexside blocks line of sight across it (§6.3). Crest is 233       /// directional and handled by the caller; here it is treated as blocking. 234       pub fn blocks_los(self) -&gt; bool { 235           matches!(self, HexsideKind::Wall   HexsideKind::Crest) 236       }</pre> |
| <a href="#">omdurman-types/src/lib.rs:443</a><br>GH:omdurman-types/src/lib.rs:443                   | Terrain::is_los_trees | <pre> 441       /// line of sight is blocked by more than two intervening tree hexes 442       /// (§6.3 note 1). 443       pub fn is_los_trees(self) -&gt; bool { 444           matches!(self, Terrain::Trees) 445       }</pre>                                                                                   |

Covered by tests: [clear\\_terrain\\_no\\_bonus](#) [building\\_gives\\_minus\\_3](#) [palm\\_grove\\_gives\\_minus\\_2](#) [rough\\_movement\\_and\\_defense](#) [swamp\\_movement\\_and\\_defense](#) [hilltop\\_movement\\_and\\_defense](#) [huts\\_movement\\_and\\_defense](#) [defense\\_modifier\\_convenience\\_matches\\_chart](#)

## §6.24 -- Anglo-Egyptian direct fire accuracy bonus and brigade integrity

implemented

manual page unknown

All Anglo-Egyptian direct fire attacks receive a +1 modifier to their die roll as an accuracy bonus. In addition, any stack of Anglo-Egyptian infantry having brigade integrity (see 5.54) receives a +1 modifier to their die roll if all four fire at the same enemy-occupied hex. These modifiers are cumulative.

See also: §5.54

| File                                                                              | Symbol                  | Code Snippet                                                                                                                                                                                                                                                                                                                                            |
|-----------------------------------------------------------------------------------|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/lib.rs:865</a><br>GH:omdurman-rules/src/lib.rs:865 | AngloEgyptianDirectFire | <pre>863   pub enum FireModifier { 864       /// +1 to all Anglo-Egyptian *direct* fire (§6.24). 865       AngloEgyptianDirectFire, 866       /// +1 brigade integrity, applied only if all four       battalions fire at 867       /// the same enemy-occupied hex (§5.54, §6.24).</pre>                                                               |
| <a href="#">omdurman-rules/src/lib.rs:850</a><br>GH:omdurman-rules/src/lib.rs:850 | BrigadeIntegrity        | <pre>848   /// stack contains all four battalions of a single       Anglo-Egyptian brigade. 849   #[derive(Serialize, Deserialize, Clone, Copy,       PartialEq, Eq, Debug)] 850   pub enum BrigadeIntegrity { 851       None, 852       Integrated(BrigadeId),</pre>                                                                                   |
| <a href="#">omdurman-rules/src/lib.rs:881</a><br>GH:omdurman-rules/src/lib.rs:881 | die_modifier            | <pre>879   impl FireModifier { 880       /// Return the numeric die-roll modifier for this       bonus/penalty (rulebook §6.24, §5.54, §6.23, §9.231, §9.232). 881       pub fn die_modifier(self) -&gt; i16 { 882           match self { 883               FireModifier::AngloEgyptianDirectFire         FireModifier::BrigadeIntegrity =&gt; 1,</pre> |

## §6.41 -- Direct Fire Subphase

implemented

manual page unknown

| File                                                                              | Symbol     | Code Snippet                                                                                                                                                                                                                                                                           |
|-----------------------------------------------------------------------------------|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/lib.rs:303</a><br>GH:omdurman-rules/src/lib.rs:303 | DirectFire | <pre>301   pub enum FireSubPhase { 302       /// Direct fire (§6.41). Both sides participate in       this sub-phase. 303       DirectFire, 304       /// Anglo-Egyptian only: Maxim second fire + named-       gunboat howitzer fire (§6.42). 305       MaximSecondAndHowitzer,</pre> |

## §6.42 -- Maxim Second Fire and Howitzer Fire Subphase

implemented

manual page unknown

| File                                                                              | Symbol                 | Code Snippet                                                                                                                                                                        |
|-----------------------------------------------------------------------------------|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/lib.rs:305</a><br>GH:omdurman-rules/src/lib.rs:305 | MaximSecondAndHowitzer | <pre>303       DirectFire, 304       /// Anglo-Egyptian only: Maxim second fire + named-       gunboat howitzer fire (§6.42). 305       MaximSecondAndHowitzer, 306   } 307  </pre> |
| <a href="#">omdurman-types/src/</a>                                               | fires_twice            | <pre>688       /// again in the Maxim Second Fire Subphase       (rulebook §6.42). The counter</pre>                                                                                |

|                                                |  |                                                                                                                                                                                        |
|------------------------------------------------|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| lib.rs:690<br>GH:omdurman-types/src/lib.rs:690 |  | <pre> 689       /// is marked "x2" in the editor to surface this. 690       pub fn fires_twice(self) -&gt; bool { 691           matches!(self, UnitFormKind::Maxim) 692       } </pre> |
|------------------------------------------------|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Covered by tests: [howitzer\\_on\\_target\\_7\\_to\\_10](#) [howitzer\\_scatters\\_below\\_7](#) [howitzer\\_short\\_on\\_5\\_6](#) [howitzer\\_long\\_on\\_3\\_4](#) [howitzer\\_left\\_right\\_on\\_1\\_2](#)

## §6.51 -- Leader Units

implemented

manual page unknown

| File                                                              | Symbol             | Code Snippet                                                                                                                                                                                                                                                                                                                                          |
|-------------------------------------------------------------------|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| omdurman-rules/src/lib.rs:408<br>GH:omdurman-rules/src/lib.rs:408 | BritishLeader      | <pre> 406       /// to claim the Mahdi's Tomb (§9.14). 407       #[derive(Serialize, Deserialize, Clone, Copy, 408               PartialEq, Eq, Hash, Debug, strum::Display)] 409       pub enum BritishLeader { 410           Kitchener, 411           Gatacre, </pre>                                                                               |
| omdurman-rules/src/lib.rs:473<br>GH:omdurman-rules/src/lib.rs:473 | BritishLeaderUnit  | <pre> 471           DervishLeaderUnit, 472           /// Anglo-Egyptian leader: movement only (§6.51). 473           BritishLeaderUnit, 474       } 475   </pre>                                                                                                                                                                                      |
| omdurman-types/src/lib.rs:683<br>GH:omdurman-types/src/lib.rs:683 | has_combat_factors | <pre> 681       /// British and Dervish leaders print a movement 682       factor only (§6.51); other 683       /// playable kinds carry fire and/or melee factors. 684       pub fn has_combat_factors(self) -&gt; bool { 685           !matches!(self, UnitFormKind::BritishLeader   686                   UnitFormKind::Marker) 687       } </pre> |

## §6.52 -- Anglo-Egyptian Friendlies Brigade

implemented

manual page unknown

| File                                                              | Symbol      | Code Snippet                                                                                                                                                                                                                                           |
|-------------------------------------------------------------------|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| omdurman-rules/src/lib.rs:636<br>GH:omdurman-rules/src/lib.rs:636 | is_friendly | <pre> 634       /// "Friendlies" units obey several special rules 635       (§5.21, §5.23, §6.52, 636       /// §9.14 victory conditions). 637       pub fn is_friendly(&amp;self) -&gt; bool { 638           matches!( 639               self, </pre> |
| omdurman-types/src/lib.rs:721<br>GH:omdurman-types/src/lib.rs:721 | Friendly    | <pre> 719       /// Native volunteer brigade -- the Shaggyeh 720       (§6.52). Do not receive 721       /// brigade integrity (§5.54 enumerates only 722       British/Egyptian/Sudanese). 723       Friendly, </pre>                                 |

## §6.53 -- Royal Engineers demolition

implemented

manual page unknown

| File | Symbol | Code Snippet |
|------|--------|--------------|
|------|--------|--------------|

|                                                                                             |                  |                                                                                                                                                                                                                                                                                                                                                                                                                  |
|---------------------------------------------------------------------------------------------|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/lib.rs:605</a><br>GH:omdurman-rules/src/lib.rs:605           | RoyalEngineers   | <pre> 603       /// The Royal Engineers (§6.53) -- a *specific*       unit, not a class, so we 604       /// model it explicitly. 605       RoyalEngineers, 606   } 607   </pre>                                                                                                                                                                                                                                 |
| <a href="#">omdurman-rules/src/lib.rs:763</a><br>GH:omdurman-rules/src/lib.rs:763           | demolishing      | <pre> 761       /// Set when the Royal Engineers are committed to a       demolition this turn 762       /// (§6.53) -- neither offensive fire nor melee       allowed that turn. 763       pub demolishing: bool, 764       /// Set when a gunboat has lost its engines to a       river mine (§10.12, roll 765       /// 5-7): it may no longer move under power and       instead drifts two hexes per </pre> |
| <a href="#">omdurman-rules/src/effects.rs:113</a><br>GH:omdurman-rules/src/effects.rs:113   | Demolition       | <pre> 111   112       /// Royal Engineers demolition (rulebook §6.53). 113       Demolition { 114           unit_id: UnitId, 115           target: DemolitionTarget, </pre>                                                                                                                                                                                                                                      |
| <a href="#">omdurman-rules/src/effects.rs:2530</a><br>GH:omdurman-rules/src/effects.rs:2530 | apply_demolition | <pre> 2528       /// resolution happens at end of turn via       [`apply_resolve_demolition`], which 2529       /// checks the engineer is still adjacent and       undisrupted. 2530       pub fn apply_demolition( 2531           state: &amp;mut GameState, 2532           unit_id: UnitId, </pre>                                                                                                            |
| <a href="#">omdurman-rules/src/lib.rs:1006</a><br>GH:omdurman-rules/src/lib.rs:1006         | DemolitionTarget | <pre> 1004       /// disrupted or driven off. 1005       #[derive(Serialize, Deserialize, Clone, Copy,       PartialEq, Eq, Debug)] 1006       pub enum DemolitionTarget { 1007           Fort(UnitId), 1008           WallHexside(HexsideRef), </pre>                                                                                                                                                           |

Covered by tests: [demolition\\_cancelled\\_when\\_engineer\\_disrupted](#) [demolition\\_cancelled\\_when\\_engineer\\_moved\\_away](#)

## §6.54 -- Forts

[implemented](#)

manual page unknown

| File                                                                              | Symbol             | Code Snippet                                                                                                                                                                                                                                                                                                                                                                                            |
|-----------------------------------------------------------------------------------|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/lib.rs:812</a><br>GH:omdurman-rules/src/lib.rs:812 | ZocReason          | <pre> 810       /// Used by the engine when answering "is this hex in       an enemy ZOC?". 811       #[derive(Serialize, Deserialize, Clone, Copy,       PartialEq, Eq, Debug)] 812       pub enum ZocReason { 813           /// Normal ZOC: any non-disrupted unit other than       an Anglo-Egyptian 814           /// leader (§5.41) projects ZOC into each of its       six adjacent hexes. </pre> |
| <a href="#">omdurman-rules/src/lib.rs:469</a><br>GH:omdurman-rules/src/lib.rs:469 | Fort               | <pre> 467           Gunboat, 468           /// Permanent emplacement -- may not move once       placed (§5.25). 469           Fort, 470           /// Dervish leader: has fire/melee/movement factors       and may melee attack. 471           DervishLeaderUnit, </pre>                                                                                                                               |
| <a href="#">omdurman-rules/src/lib.rs:773</a><br>GH:omdurman-rules/src/lib.rs:773 | UnitState::may_act | <pre> 771       impl UnitState { 772           /// A disrupted unit may not move, fire, or melee       (rulebook §5, reference notes). 773           pub fn may_act(self) -&gt; bool { </pre>                                                                                                                                                                                                           |



|                                                                                   |                          |                                                                                                                                                                                                                                                                                                                                                              |
|-----------------------------------------------------------------------------------|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                   |                          | <pre> 774           !self.disrupted 775       } </pre>                                                                                                                                                                                                                                                                                                       |
| <a href="#">omdurman-rules/src/lib.rs:751</a><br>GH:omdurman-rules/src/lib.rs:751 | UnitState                | <pre> 749       /// rather than one big enum. 750       #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug, Default)] 751       pub struct UnitState { 752           /// Reference table: "Disrupted units: no ZOC; may not move; may not fire 753           /// offensively or defensively; may not melee; are turned face up at the </pre> |
| <a href="#">omdurman-rules/src/lib.rs:910</a><br>GH:omdurman-rules/src/lib.rs:910 | FireAttack               | <pre> 908       /// modifiers (rulebook §6). 909       #[derive(Serialize, Deserialize, Clone, Debug)] 910       pub struct FireAttack { 911           pub firing_player: Player, 912           pub phase: Phase, </pre>                                                                                                                                     |
| <a href="#">omdurman-rules/src/lib.rs:925</a><br>GH:omdurman-rules/src/lib.rs:925 | FireAttack::net_modifier | <pre> 923       impl FireAttack { 924           /// Sum of all fire modifiers applied to this attack (rulebook §6.24). 925           pub fn net_modifier(&amp;self) -&gt; i16 { 926               self.modifiers.iter().map( m  m.die_modifier()).sum() 927           } </pre>                                                                               |

## §6.61 -- Only artillery may fire at gunboats; 3+ to sink

implemented

manual page unknown

Only artillery may fire at gunboats. A result of 3 or more on the combat results table is required to sink a gunboat. Any other result is a miss.

| File                                                                              | Symbol      | Code Snippet                                                                                                                                                                                                                                                                                                                        |
|-----------------------------------------------------------------------------------|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/lib.rs:502</a><br>GH:omdurman-rules/src/lib.rs:502 | WeaponClass | <pre> 500       /// enum so a "spear" unit cannot accidentally fire on the "Howitzer" line. 501       #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug, strum::Display)] 502       pub enum WeaponClass { 503           /// Dervish spears and swords -- no ranged fire at all. 504           Melee, </pre> |

Covered by tests: [rifles\\_may\\_not\\_sink\\_a\\_gunboat](#)

## §6.62 -- Only artillery may fire at forts; 2+ to destroy

implemented

manual page unknown

Only artillery may fire at forts. A result of 2 or more on the combat results table is required to eliminate a fort. Any other result is a miss. If the fort contains any enemy units at the instant it is destroyed, one unit is eliminated with the fort.

| File                                                                              | Symbol      | Code Snippet                                                                                                                                                                                                                                                                                              |
|-----------------------------------------------------------------------------------|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/lib.rs:502</a><br>GH:omdurman-rules/src/lib.rs:502 | WeaponClass | <pre> 500       /// enum so a "spear" unit cannot accidentally fire on the "Howitzer" line. 501       #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug, strum::Display)] 502       pub enum WeaponClass { 503           /// Dervish spears and swords -- no ranged fire at </pre> |

|  |  |                      |
|--|--|----------------------|
|  |  | all.<br>504   Melee, |
|--|--|----------------------|

## §6.63 -- Only artillery may breach wall hexsides; 2+ required

implemented  
manual page unknown

Only artillery may fire to breach a wall hexside of Khartoum or the walled city of Omdurman. A result of 2 or more on the combat results table is required to breach a wall. Any other result is a miss. The effect of the breach is to negate the wall hexside for line of sight purposes. Place a "BREACH" marker in an adjacent hex so that the arrow points to the breached hexside. If any enemy units are adjacent to the wall hexside at the instant it is breached, one enemy unit is eliminated.

**\*\*6.64) Howitzer fire:\*\*** Five units in the game have howitzer fire capability. These are the five named British gunboats. They may fire their artillery factor as direct fire during the Direct Fire Subphase (see 4 and 6.41) and may then fire the same artillery factor as howitzer fire during the Maxim Second Fire and Howitzer Subphase (see 4 and 6.42). Exception: no howitzer fire is allowed during night game turns. To fire howitzer fire, select any target hex between 4 and 10 hexes from the firing gunboat (ignoring the Line of Sight Table) and roll the ten-sided die twice. The first die roll is the Combat Results Table die roll and the second roll is the impact hex die roll. Refer to the Howitzer Fire Scattergram on the marchpost for the impact hex. The designated target hex is hit on a roll of 7-10. Once a howitzer

(see manual for full text)

See also: §6.41, §6.42, §6.64

| File                                                                              | Symbol | Code Snippet                                                                                                                                                                                                                                                                                                                       |
|-----------------------------------------------------------------------------------|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-types/src/lib.rs:204</a><br>GH:omdurman-types/src/lib.rs:204 | Breach | <pre> 202       /// Breach in a wall (artillery/§6.63 or Royal           Engineers/§6.53). ZOC both 203       /// ways; LOS no longer blocked across the hexside. 204       Breach, 205       /// Khor -- gully/wadi. ZOCs do not extend across           (§5.44); advance after 206       /// combat may not cross (§6.82).</pre> |

## §6.64 -- Howitzer fire

implemented  
manual page unknown

| File                                                                                                      | Symbol           | Code Snippet                                                                                                                                                                                                                                                                                                                                               |
|-----------------------------------------------------------------------------------------------------------|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/howitzer_scatter.rs:6</a><br>GH:omdurman-rules/src/howitzer_scatter.rs:6   | ScatterDirection | <pre> 4       /// (§6.64). The caller maps these to hex-grid offsets. 5       #[derive(Clone, Copy, PartialEq, Eq, Debug)] 6       pub enum ScatterDirection { 7           /// Roll 7-10: hit the target hex. 8           OnTarget,</pre>                                                                                                                  |
| <a href="#">omdurman-rules/src/howitzer_scatter.rs:28</a><br>GH:omdurman-rules/src/howitzer_scatter.rs:28 | howitzer_scatter | <pre> 26       ///   3-4   [`ScatterDirection::Long`] (upstream)   27       ///   1-2   [`ScatterDirection::LeftRight`]   28       pub fn howitzer_scatter(impact_roll: DieRoll) -&gt;           ScatterDirection { 29           use DieRoll::*; 30           match impact_roll {</pre>                                                                    |
| <a href="#">omdurman-rules/src/lib.rs:419</a><br>GH:omdurman-rules/src/lib.rs:419                         | GunboatId        | <pre> 417       /// fire; "old" gunboats do not (rulebook §2.32). 418       #[derive(Serialize, Deserialize, Clone, Copy,           PartialEq, Eq, Hash, Debug, strum::Display)] 419       pub enum GunboatId { 420           /// One of the five new-type named gunboats with               howitzer capability. 421           Named(NamedGunboat),</pre> |
| <a href="#">omdurman-rules/src/effects.rs:62</a>                                                          | HowitzerFire     | <pre> 60   61       /// Resolve a howitzer bombardment (two rolls:           Combat Results Table + impact scatter) (rulebook §6.64).</pre>                                                                                                                                                                                                                |

|                                                                             |                                    |                                                                                                                                                                                                                                                                                                                                        |
|-----------------------------------------------------------------------------|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GH:omdurman-rules/src/effects.rs:62                                         |                                    | <pre> 62       HowitzerFire { 63           attack: FireAttack, 64           combat_results_table_roll: DieRoll, </pre>                                                                                                                                                                                                                 |
| omdurman-rules/src/effects.rs:1912<br>GH:omdurman-rules/src/effects.rs:1912 | apply_howitzer_fire                | <pre> 1910   1911       /// Validate and apply a howitzer fire attack (scatter 1912       path) (rulebook §6.64). 1913       pub fn apply_howitzer_fire( 1914           state: &amp;mut GameState, 1915           attack: &amp;FireAttack, </pre>                                                                                      |
| omdurman-rules/src/lib.rs:946<br>GH:omdurman-rules/src/lib.rs:946           | HowitzerResolution                 | <pre> 944       /// roll on the Howitzer Fire Scattergram (§6.64). 945       #[derive(Serialize, Deserialize, Clone, Copy, 946               PartialEq, Eq, Debug)] 947       pub struct HowitzerResolution { 948           pub combat_results_table_roll: DieRoll, 949           pub impact_roll: DieRoll, </pre>                     |
| omdurman-rules/src/lib.rs:953<br>GH:omdurman-rules/src/lib.rs:953           | HowitzerResolution::hit_target_hex | <pre> 951       impl HowitzerResolution { 952           /// The designated target hex is hit on impact roll 953           7-10 (§6.64). 954           pub fn hit_target_hex(self) -&gt; bool { 955               use DieRoll::*; 956               matches!(self.impact_roll, Seven   Eight   Nine 957                     Ten) </pre> |
| omdurman-rules/src/effects.rs:975<br>GH:omdurman-rules/src/effects.rs:975   | can_fire_at                        | <pre> 973           /// modifier in the [`FireAttack`] and is 974           responsible for the LOS gate. 975           /// (Howitzer fire ignores LOS entirely -- §6.64.) 976           pub fn can_fire_at( 977               &amp;self, 978               firer: UnitId, </pre>                                                      |

Covered by tests: [howitzer\\_on\\_target\\_7\\_to\\_10](#) [howitzer\\_scatters\\_below\\_7](#) [howitzer\\_short\\_on\\_5\\_6](#) [howitzer\\_long\\_on\\_3\\_4](#) [howitzer\\_left\\_right\\_on\\_1\\_2](#)

## §6.81 -- Moving player may fire with all capable units

implicit

manual page unknown

During Offensive Fire phase, the moving player may fire with all of his units capable of firing, up to their maximum range, within the limitations imposed by the rules of combat.

## §6.82 -- Advance after combat (offensive fire)

implemented

manual page unknown

If an enemy-occupied hex is vacated as a result of offensive fire, friendly units may advance after combat into the vacated hex. To be eligible to advance, the friendly units must have participated in the attack and must have been adjacent to the vacated hex. Note that artillery may not advance, nor may units advance across a wall hexside (except at a gate or breach). Units may never advance after combat across a khor.

| File                                                                      | Symbol             | Code Snippet                                                                                                                                                                                                                                                                                                            |
|---------------------------------------------------------------------------|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| omdurman-rules/src/effects.rs:100<br>GH:omdurman-rules/src/effects.rs:100 | AdvanceAfterCombat | <pre> 98       /// after fire, §7.6 after melee). Eligible units 99       are adjacent attackers 100       /// that are not artillery; the target hex must be 101       empty of enemies. 102       AdvanceAfterCombat { unit_id: UnitId, to: 103           HexCoord }, 104   105       // -- Unit state changes </pre> |

|                                                                                             |                           |                                                                                                                                                                                                                                                                                                                                                                                 |
|---------------------------------------------------------------------------------------------|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| omdurman-<br>rules/src/<br>effects.rs:2478<br>GH:omdurman-<br>rules/src/<br>effects.rs:2478 | apply_advance_after_comb  | 2476  <br>2477   /// Apply an advance-after-combat for a unit (rulebook<br>\$6.82, \$7.6).<br>2478   pub fn apply_advance_after_combat(<br>2479       state: &mut GameState,<br>2480       unit_id: UnitId,                                                                                                                                                                     |
| omdurman-<br>rules/src/<br>effects.rs:100<br>GH:omdurman-<br>rules/src/<br>effects.rs:100   | AdvanceAfterCombat        | 98       /// after fire, \$7.6 after melee). Eligible units<br>are adjacent attackers<br>99       /// that are not artillery; the target hex must be<br>empty of enemies.<br>100       AdvanceAfterCombat { unit_id: UnitId, to:<br>HexCoord },<br>101  <br>102       // -- Unit state changes<br>-----                                                                         |
| omdurman-<br>types/src/<br>lib.rs:245<br>GH:omdurman-<br>types/src/lib.rs:245               | blocks_advance_after_comb | 243  <br>244       /// Whether advance-after-combat may *not* cross<br>this side (\$6.82, \$7.6).<br>245       pub fn blocks_advance_after_combat(self) -> bool {<br>246           matches!(<br>247               self,                                                                                                                                                         |
| omdurman-<br>rules/src/<br>effects.rs:2374<br>GH:omdurman-<br>rules/src/<br>effects.rs:2374 | can_advance_after_combat  | 2372       /// player's unit, not artillery, adjacent to<br>`to`, and `to` now empty.<br>2373       /// Wall/khor hexside restrictions are not<br>enforced (no hexside map data).<br>2374       pub fn can_advance_after_combat(&self, unit_id:<br>UnitId, to: HexCoord) -> Result<(), RuleError> {<br>2375           let unit = self<br>2376               .find_unit(unit_id) |

7/7 implemented

## §7 -- Melee Phase

### §7.1 -- Melee strength printed on counter

implemented

manual page unknown

The melee strength of all units is printed on the counter. Note that gunboats have no melee strength. Gunboats may neither melee attack nor be melee attacked.

| File                                                                              | Symbol                        | Code Snippet                                                                                                                                                                                                                                                                              |
|-----------------------------------------------------------------------------------|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/lib.rs:91</a><br>GH:omdurman-rules/src/lib.rs:91   | <b>MeleeFactor</b>            | <pre>89   // Every possible value from the annotated counter       set is a named variant. 90   #[derive(Serialize, Deserialize, Clone, Copy,       PartialEq, Eq, PartialOrd, Ord, Hash, Debug, strum::Display)] 91   pub enum MeleeFactor { 92       One = 1, 93       Three = 3,</pre> |
| <a href="#">omdurman-rules/src/lib.rs:102</a><br>GH:omdurman-rules/src/lib.rs:102 | <b>MeleeFactor::sum</b>       | <pre>100   impl MeleeFactor { 101       // Sum multiple melee factors (rulebook §7.1). 102       pub fn sum&lt;'a&gt;(factors: impl IntoIterator&lt;Item =       &amp;'a MeleeFactor&gt;) -&gt; u16 { 103           factors.into_iter().map( f  f.value()).sum() 104       }</pre>        |
| <a href="#">omdurman-rules/src/lib.rs:487</a><br>GH:omdurman-rules/src/lib.rs:487 | <b>may_be_melee_attacked</b>  | <pre>485   486   // Gunboats neither attack nor are attacked in       melee (§7.1). 487   pub fn may_be_melee_attacked(self) -&gt; bool { 488       !matches!(self, UnitKind::Gunboat) 489   }</pre>                                                                                      |
| <a href="#">omdurman-rules/src/lib.rs:487</a><br>GH:omdurman-rules/src/lib.rs:487 | <b>UnitKind::may_be_melee</b> | <pre>485   486   // Gunboats neither attack nor are attacked in       melee (§7.1). 487   pub fn may_be_melee_attacked(self) -&gt; bool { 488       !matches!(self, UnitKind::Gunboat) 489   }</pre>                                                                                      |

### §7.2 -- Melee adjacent only, not across wall hexsides

implemented

manual page unknown

Melee simulates the hand-to-hand fighting of the period. Units may melee attack adjacent enemy units only. Units may not melee attack across a wall hexside, but may melee attack through a gate or breach hexside.

| File                                                                                        | Symbol              | Code Snippet                                                                                                                                                                                                                                                                                            |
|---------------------------------------------------------------------------------------------|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-types/src/lib.rs:240</a><br>GH:omdurman-types/src/lib.rs:240           | <b>blocks_melee</b> | <pre>238   // Whether melee may *not* be made across this       side (§7.2). Gates and 239   // breaches are passable to melee. 240   pub fn blocks_melee(self) -&gt; bool { 241       matches!(self, HexsideKind::Wall         HexsideKind::ZaribaThornHedge) 242   }</pre>                            |
| <a href="#">omdurman-rules/src/effects.rs:1109</a><br>GH:omdurman-rules/src/effects.rs:1109 | <b>can_melee</b>    | <pre>1107   // Does **not** check wall/khor hexsides (§7.2)       -- those need the game map, 1108   // which the rules engine does not hold; the app       gates on them. 1109   pub fn can_melee(&amp;self, attacker: UnitId,       defender_hex: HexCoord) -&gt; Result&lt;(), RuleError&gt; {</pre> |

|  |  |                                                                                 |
|--|--|---------------------------------------------------------------------------------|
|  |  | <pre> 1110           let unit = self 1111           .find_unit(attacker) </pre> |
|--|--|---------------------------------------------------------------------------------|

### §7.3 -- Simultaneous melee combat

implemented

manual page unknown

Melee combat is considered simultaneous, so that units eliminated by melee attacks still get a melee combat die roll.

| File                                                                                        | Symbol             | Code Snippet                                                                                                                                                                                                  |
|---------------------------------------------------------------------------------------------|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/effects.rs:72</a><br>GH:omdurman-rules/src/effects.rs:72     | MeleeCombat        | <pre> 70       /// Used for an immediate resolution with no 71       /// resolution primitive in tests). 72       MeleeCombat { 73           attack: MeleeAttack, 74           attacker_roll: DieRoll, </pre> |
| <a href="#">omdurman-rules/src/effects.rs:2110</a><br>GH:omdurman-rules/src/effects.rs:2110 | apply_melee_combat | <pre> 2108   2109       /// Apply a simultaneous melee combat between two 2110       pub fn apply_melee_combat( 2111           state: &amp;mut GameState, 2112           attack: &amp;MeleeAttack, </pre>     |

### §7.4 -- Who may melee attack / defend

implemented

manual page unknown

Only infantry, cavalry, camel units, and Dervish leaders may melee attack. All units (except gunboats — see 7.1) may melee defend.

See also: §7.1

| File                                                                              | Symbol           | Code Snippet                                                                                                                                                                                                                                                                                                                                            |
|-----------------------------------------------------------------------------------|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/lib.rs:479</a><br>GH:omdurman-rules/src/lib.rs:479 | may_melee_attack | <pre> 477       /// Rulebook §7.4 -- only infantry, cavalry, camel 478       /// and Dervish leaders may 479       /// melee *attack*. All others (except gunboats) 480       /// may melee *defend* (§7.1). 481       pub fn may_melee_attack(self) -&gt; bool { 482           matches!( 483               self, </pre>                                |
| <a href="#">omdurman-rules/src/lib.rs:459</a><br>GH:omdurman-rules/src/lib.rs:459 | UnitKind         | <pre> 457       /// engine prove the constraint. 458       #[derive(Serialize, Deserialize, Clone, Copy, 459               PartialEq, Eq, Hash, Debug)] 460       pub enum UnitKind { 461           /// Foot infantry. Includes Anglo-Egyptian 462           infantry, "Friendlies", 463           /// Royal Engineers, and Dervish foot tribes. </pre> |
| <a href="#">omdurman-rules/src/lib.rs:335</a><br>GH:omdurman-rules/src/lib.rs:335 | DervishTribe     | <pre> 333       // 334       ----- 335       pub use omdurman_types::DervishTribe; 336   337       /// Anglo-Egyptian infantry brigades -- designation 338       printed on the counter </pre>                                                                                                                                                          |
| <a href="#">omdurman-rules/src/effects.rs:1109</a>                                | can_melee        | <pre> 1107       /// Does <b>not</b> check wall/khor hexsides (§7.2) 1108       -- those need the game map, 1109       /// which the rules engine does not hold; the app </pre>                                                                                                                                                                         |

|                                       |  |                                                                                                                                                                                                                                     |
|---------------------------------------|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GH:omdurman-rules/src/effects.rs:1109 |  | <pre> gates on them. 1109           pub fn can_melee(&amp;self, attacker: UnitId, defender_hex: HexCoord) -&gt; Result&lt;(), RuleError&gt; { 1110               let unit = self 1111                   .find_unit(attacker) </pre> |
|---------------------------------------|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## \$7.5 -- Cavalry/camel retreat before melee

implemented

manual page unknown

Cavalry and camel units may retreat two hexes from an infantry melee attack. Note, however, that only one retreat per unit per turn is permitted. Thus, if their retreat places them adjacent to enemy units whose melee attacks have not yet been resolved, those enemy units may elect to attack the retreating unit(s).

| File                                                                        | Symbol                     | Code Snippet                                                                                                                                                                                                                                                                                                                                                                                 |
|-----------------------------------------------------------------------------|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| omdurman-rules/src/effects.rs:95<br>GH:omdurman-rules/src/effects.rs:95     | RetreatBeforeMelee         | <pre> 93           /// melee attack, *before* it is resolved (§7.5). One retreat per unit per 94           /// turn. (rulebook §7.5). 95           RetreatBeforeMelee { unit_id: UnitId, to: HexCoord }, 96   97           /// An attacking unit advances into a hex vacated by combat (rulebook §6.82 </pre>                                                                                |
| omdurman-rules/src/effects.rs:2463<br>GH:omdurman-rules/src/effects.rs:2463 | apply_retreat_before_melee | <pre> 2461   2462           /// Apply a retreat-before-melee for a cavalry/camel unit (rulebook §7.5). 2463           pub fn apply_retreat_before_melee( 2464               state: &amp;mut GameState, 2465               unit_id: UnitId, </pre>                                                                                                                                            |
| omdurman-rules/src/lib.rs:493<br>GH:omdurman-rules/src/lib.rs:493           | may_retreat_before_melee   | <pre> 491           /// Cavalry and camel units may retreat two hexes from an infantry melee 492           /// attack (§7.5). 493           pub fn may_retreat_before_melee(self) -&gt; bool { 494               matches!(self, UnitKind::Cavalry   UnitKind::Camel) 495           } </pre>                                                                                                  |
| omdurman-rules/src/lib.rs:157<br>GH:omdurman-rules/src/lib.rs:157           | HexDistance                | <pre> 155           /// (rulebook §6.22, §7.5). 156           #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, PartialOrd, Ord, Hash, Debug)] 157           pub struct HexDistance(pub u16); 158   159           impl HexDistance { </pre>                                                                                                                                       |
| omdurman-rules/src/effects.rs:2329<br>GH:omdurman-rules/src/effects.rs:2329 | can_retreat_before_melee   | <pre> 2327           /// two hexes away and empty. (Does not verify the attacker is infantry -- 2328           /// the caller offers the retreat only in response to one.) 2329           pub fn can_retreat_before_melee(&amp;self, unit_id: UnitId, to: HexCoord) -&gt; Result&lt;(), RuleError&gt; { 2330               let unit = self 2331                   .find_unit(unit_id) </pre> |

Covered by tests: retreat\_before\_melee\_only\_cavalry\_two\_hexes

## \$7.6 -- Advance after melee

implemented

manual page unknown

If a melee attack eliminates all of the defenders in an adjacent hex, the Dervish player MUST advance into the vacated hex. To be eligible to advance, the Dervish units must have been adjacent to the vacated hex and participated in the

melee attack that eliminated the defenders. All surviving eligible Dervish units **MUST** advance, up to the stacking limit. The Anglo-Egyptian player may advance if desired. Note that only attacking units may advance.

| File                                                                                        | Symbol                            | Code Snippet                                                                                                                                                                                                                                                                                                                                                                                                    |
|---------------------------------------------------------------------------------------------|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/effects.rs:100</a><br>GH:omdurman-rules/src/effects.rs:100   | <b>AdvanceAfterCombat</b>         | <pre> 98       /// after fire, §7.6 after melee). Eligible units          are adjacent attackers 99       /// that are not artillery; the target hex must be          empty of enemies. 100       AdvanceAfterCombat { unit_id: UnitId, to:          HexCoord }, 101   102       // -- Unit state changes          ----- </pre>                                                                                 |
| <a href="#">omdurman-rules/src/effects.rs:2478</a><br>GH:omdurman-rules/src/effects.rs:2478 | <b>apply_advance_after_combat</b> | <pre> 2476   2477       /// Apply an advance-after-combat for a unit (rulebook          §6.82, §7.6). 2478       pub fn apply_advance_after_combat(          state: &amp;mut GameState, 2479       unit_id: UnitId, 2480   </pre>                                                                                                                                                                               |
| <a href="#">omdurman-rules/src/effects.rs:2374</a><br>GH:omdurman-rules/src/effects.rs:2374 | <b>can_advance_after_combat</b>   | <pre> 2372       /// player's unit, not artillery, adjacent to          `to`, and `to` now empty. 2373       /// Wall/khor hexside restrictions are not          enforced (no hexside map data). 2374       pub fn can_advance_after_combat(&amp;self, unit_id:          UnitId, to: HexCoord) -&gt; Result&lt;(), RuleError&gt; { 2375           let unit = self 2376               .find_unit(unit_id) </pre> |

## §7.7 -- Melee modifiers

implemented

manual page unknown

To resolve melee, both the attacker and the defender roll on the Combat Results Table and apply the applicable melee modifier to their die roll. The Dervish player receives a +2 melee modifier, the Anglo-Egyptian player receives a +1 melee modifier. No terrain modifiers are applied to melee combat (Exception: Zariba hexsides in the historical scenario and the campaign game, if constructed — see 9.23). Melee losses must be taken from meleeing units first!

See also: §9.23

| File                                                                              | Symbol               | Code Snippet                                                                                                                                                                                                                                                                                               |
|-----------------------------------------------------------------------------------|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/lib.rs:964</a><br>GH:omdurman-rules/src/lib.rs:964 | <b>MeleeModifier</b> | <pre> 962   963       #[derive(Serialize, Deserialize, Clone, Copy,          PartialEq, Eq, Debug)] 964       pub enum MeleeModifier {          /// +2 to all Dervish melee rolls (§7.7). 965       DervishStandard, 966   </pre>                                                                          |
| <a href="#">omdurman-rules/src/lib.rs:198</a><br>GH:omdurman-rules/src/lib.rs:198 | <b>DieModifier</b>   | <pre> 196       /// A die-roll modifier from a single named source          (rulebook §6.24, §7.7). 197       #[derive(Serialize, Deserialize, Clone, Copy,          PartialEq, Eq, Debug, Default)] 198       pub enum DieModifier {          #[default] 199       Zero, 200   </pre>                     |
| <a href="#">omdurman-rules/src/lib.rs:987</a><br>GH:omdurman-rules/src/lib.rs:987 | <b>MeleeAttack</b>   | <pre> 985       /// A melee attack: simultaneous, both sides roll on          the Combat Results Table (§7.3, §7.7). 986       #[derive(Serialize, Deserialize, Clone, Debug)] 987       pub struct MeleeAttack {          pub attacker_player: Player, 988       pub attacker_hex: HexCoord, 989   </pre> |



|                                                                                   |                                               |                                                                                                                                                                                                                                                                         |
|-----------------------------------------------------------------------------------|-----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/lib.rs:968</a><br>GH:omdurman-rules/src/lib.rs:968 | <b>MeleeModifier::AngloEgyptianStandard</b>   | <pre> 966       DervishStandard, 967       /// +1 to all Anglo-Egyptian melee rolls (§7.7). 968       AngloEgyptianStandard, 969       /// Inverted to -2 when Dervish units melee-attack     across a trench into 970       /// an entrenched defender (§9.232).</pre> |
| <a href="#">omdurman-rules/src/lib.rs:971</a><br>GH:omdurman-rules/src/lib.rs:971 | <b>MeleeModifier::DervishVsTrenchDefender</b> | <pre> 969       /// Inverted to -2 when Dervish units melee-attack     across a trench into 970       /// an entrenched defender (§9.232). 971       DervishVsTrenchedDefender, 972   } 973  </pre>                                                                     |
| <a href="#">omdurman-rules/src/lib.rs:966</a><br>GH:omdurman-rules/src/lib.rs:966 | <b>MeleeModifier::DervishStandard</b>         | <pre> 964   pub enum MeleeModifier { 965       /// +2 to all Dervish melee rolls (§7.7). 966       DervishStandard, 967       /// +1 to all Anglo-Egyptian melee rolls (§7.7). 968       AngloEgyptianStandard,</pre>                                                   |

Covered by tests: [melee\\_resolves\\_simultaneously](#)

■ 2/2 implemented

## §8 -- Night Game Turns

### §8.1 -- Night effects

implemented

manual page unknown

The effects of night game turns are: a) all Anglo-Egyptian movement allowances are halved (round down), b) there is no Anglo-Egyptian howitzer fire, and c) all fire ranges are halved for both sides (round down, but range 1 stays range 1). Range effects on fire combat are the same as during day game turns. For example, an Anglo-Egyptian infantry unit firing at night will be doubled at range 1, normal at range 2, and may not fire at range 3 or greater.

**\*\*8.2) Dervish Desertion Roll:\*\*** Once each campaign game, during the first night turn of the game, the Dervish player rolls one die to see how many of his units desert. The roll is made during the movement phase and the number of deserting Dervish units is equal to 1½ times the roll of one die. The Dervish player may choose which units desert by merely removing them from the mapsheet. The KHALIFA unit, gunboats, artillery units, and forts are the only Dervish units that may not be chosen. No victory points are awarded to the Anglo-Egyptian player for deserting Dervishes.

(see manual for full text)

See also: §8.2

| File                                                                                | Symbol                      | Code Snippet                                                                                                                                                                                                                                                                                                       |
|-------------------------------------------------------------------------------------|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/lib.rs:137</a><br>GH:omdurman-rules/src/lib.rs:137   | MovementAllowance::halve    | <pre>135   impl MovementAllowance { 136       /// Night movement allowance = halved (round down)       (rulebook §8.1, §5.11). 137       pub fn halve(self) -&gt; Self { 138           let v = self.value() / 2; 139           MovementAllowance::try_from(v).expect("halved value always a named variant")</pre>  |
| <a href="#">omdurman-rules/src/lib.rs:266</a><br>GH:omdurman-rules/src/lib.rs:266   | DayNight                    | <pre>264   /// (rulebook §8.1). 265   #[derive(Serialize, Deserialize, Clone, Copy,       PartialEq, Eq, Debug)] 266   pub enum DayNight { 267       Day, 268       Night,</pre>                                                                                                                                   |
| <a href="#">omdurman-rules/src/lib.rs:1351</a><br>GH:omdurman-rules/src/lib.rs:1351 | effective_range_at_night    | <pre>1349   /// Apply night-turn range halving (§8.1): "all fire       ranges are halved for 1350   /// both sides (round down, but range 1 stays range       1)." 1351   pub fn effective_range_at_night(range: HexDistance) -&gt;       HexDistance { 1352       if range.0 &lt;= 1 { 1353           range</pre> |
| <a href="#">omdurman-rules/src/lib.rs:1361</a><br>GH:omdurman-rules/src/lib.rs:1361 | effective_movement_at_night | <pre>1359   /// Apply night-turn movement halving for Anglo-       Egyptian units (§8.1): all 1360   /// Anglo-Egyptian movement allowances are halved       (round down). 1361   pub fn effective_movement_at_night(       allowance: MovementAllowance, 1362       player: Player,</pre>                         |

Covered by tests: [night\\_max\\_ranges](#) [night\\_max\\_ranges\\_remaining](#) [ae\\_rifle\\_at\\_night\\_matches\\_rulebook\\_example](#) [max\\_day\\_range\\_all\\_combos](#)

### §8.2 -- Dervish Desertion Roll

implemented

manual page unknown

| File | Symbol | Code Snippet |
|------|--------|--------------|
|------|--------|--------------|

|                                                                                               |                  |                                                                                                                                                                                                                                                                                                 |
|-----------------------------------------------------------------------------------------------|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/effects.rs:128</a><br>GH:omdurman-rules/src/effects.rs:128     | DervishDesertion | <pre> 126       /// the effect. The Khalifa, gunboats, artillery, and forts may not be 127       /// chosen. 128       DervishDesertion { 129           roll: DieRoll, 130           deserters: Vec&lt;UnitId&gt;, </pre>                                                                       |
| <a href="#">omdurman-rules/src/turn_track.rs:60</a><br>GH:omdurman-rules/src/turn_track.rs:60 | DervishDesertion | <pre> 58           None, 59           /// Dervish desertion roll (§8.2) -- occurs on the first night turn. 60           DervishDesertion, 61           /// Dervish reinforcements are available. 62           DervishReinforcements, </pre>                                                     |
| <a href="#">omdurman-rules/src/turn_track.rs:57</a><br>GH:omdurman-rules/src/turn_track.rs:57 | TurnEvent        | <pre> 55       /// Special events that occur on specific turns (rulebook §8.2, §9.112, §9.113). 56       #[derive(Clone, Copy, PartialEq, Eq, Debug)] 57       pub enum TurnEvent { 58           None, 59           /// Dervish desertion roll (§8.2) -- occurs on the first night turn. </pre> |

Covered by tests: [desertion\\_on\\_first\\_night](#)

 17/31 implemented

## §9 -- The Scenarios

### §9.1 -- The Campaign Game

descriptive

manual page unknown

#### The Campaign Game

### §9.2 -- The Historical Scenario

descriptive

manual page unknown

The Historical Scenario

Players should note that the historical scenario is an exercise in futility for the Dervish player. It is, however, an interesting demonstration of the absolute imbecility of the Khalifa's generalship and vividly shows the superiority of entrenched firepower over traditional tribal arms in the colonial period.

### §9.3 -- Bonus Game: Fall of Khartoum

out-of-scope

manual page unknown

#### Bonus Game: FALL OF KHARTOUM scenario

### §9.11 -- Set Up (Campaign)

out-of-scope

manual page unknown

#### Set Up

### §9.12 -- Scenario Length (Campaign)

implemented

manual page unknown

Scenario Length

6:00 am, Sept. 1 through 8:00 am, Sept. 3, 22 Game Turns.

| File                                                                                          | Symbol    | Code Snippet                                                                                                                                                                                                                                        |
|-----------------------------------------------------------------------------------------------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/turn_track.rs:8</a><br>GH:omdurman-rules/src/turn_track.rs:8   | GameTime  | <pre>6   /// starts at one of these twelve times.<br/>7   #[derive(Clone, Copy, PartialEq, Eq, Debug)]<br/>8   pub enum GameTime {<br/>9       SixAM,<br/>10       EightAM,</pre>                                                                   |
| <a href="#">omdurman-rules/src/turn_track.rs:44</a><br>GH:omdurman-rules/src/turn_track.rs:44 | TurnEntry | <pre>42   /// A single entry on the Turn Record Track (rulebook<br/>    §9.12, §9.22).<br/>43   #[derive(Clone, Copy, PartialEq, Eq, Debug)]<br/>44   pub struct TurnEntry {<br/>45       /// 1-based turn number.<br/>46       pub turn: u8,</pre> |

|                                                                                                 |                                     |                                                                                                                                                                                                                                                                                                             |
|-------------------------------------------------------------------------------------------------|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/turn_track.rs:92</a><br>GH:omdurman-rules/src/turn_track.rs:92   | <a href="#">CAMPAIGN_TURN_TRACK</a> | <pre>90   /// is turn 9, which carries the once-per-game Dervish 91   /// the printed track prints "Dervish Desertion Roll / 92   pub const CAMPAIGN_TURN_TRACK: [TurnEntry; 22] = [ 93       // Row 1, left-&gt;right: Sept 1, 6 am -&gt; 8 pm, then 94       entry(1, SixAM, Day, TurnEvent::None),</pre> |
| <a href="#">omdurman-rules/src/turn_track.rs:238</a><br>GH:omdurman-rules/src/turn_track.rs:238 | <a href="#">TurnLabel</a>           | <pre>236   /// The track is a 9 × 3 grid with a snake layout. 237   #[derive(Debug, Clone, Copy, PartialEq, Eq)] 238   pub enum TurnLabel { 239       /// A cell that has no printed label (unused 240       Blank,</pre>                                                                                   |
| <a href="#">omdurman-rules/src/turn_track.rs:299</a><br>GH:omdurman-rules/src/turn_track.rs:299 | <a href="#">turn_marker_pixel</a>   | <pre>297   /// 298   /// Rows 0 and 1 use all 9 columns; row 2 uses only 299   pub fn turn_marker_pixel(track: &amp;omdurman_types::CampaignTurnTrack, turn: u8) -&gt; (f32, f32) { 300       let cell_w = track.w / 9.0; 301       let cell_h = track.h / 3.0;</pre>                                       |

Covered by tests: [campaign\\_track\\_22\\_turns](#) [desertion\\_on\\_first\\_night](#) [campaign\\_track\\_label\\_and\\_day\\_night\\_agree](#) [game\\_time\\_display\\_all\\_variants](#) [turn\\_label\\_display](#) [turn\\_label\\_out\\_of\\_range\\_is\\_none](#) [turn\\_marker\\_pixel\\_row\\_0\\_left\\_to\\_right](#) [turn\\_marker\\_pixel\\_row\\_1\\_right\\_to\\_left](#) [turn\\_marker\\_pixel\\_rows\\_are\\_stacked](#)

§9.13 -- Special Rules (Campaign)

[descriptive](#)  
manual page unknown

Special Rules

None.

§9.14 -- Victory Conditions (Campaign)

[implemented](#)  
manual page unknown

Victory Conditions

The Mahdi’s Tomb in Omdurman was not only the tallest structure in the entire Sudan in 1898, it was also a Dervish holy shrine. Its loss or destruction would be a severe blow to the Mahdist cause. It is accordingly assigned 25 victory points which are awarded to the player who controls it at the conclusion of play. The Dervish player controls it at the start of play. As a tactical note, the Anglo-Egyptian player will find a decisive victory almost impossible unless he takes the Mahdi’s Tomb from the Dervish player. To take the Tomb hex, it must be occupied by one British leader plus any one non-“Friendlies” Anglo-Egyptian combat unit (both undisrupted) at the conclusion of play.

Additional victory points are awarded as follows:

**\*\*Dervish Player receives:\*\***

- 10 pts: each British leader eliminated

(see manual for full text)

See also: §5.21

| File                                                           | Symbol                   | Code Snippet                                                                                                                                                                         |
|----------------------------------------------------------------|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/lib.rs:1090</a><br>GH:omdurman- | <a href="#">VpSource</a> | <pre>1088   /// the manual and the engine. 1089   #[derive(Serialize, Deserialize, Clone, Copy, 1090   pub enum VpSource { 1091       // ----- Anglo-Egyptian player receives:</pre> |

|                                                                                             |                            |                                                                                                                                                                                                                                                                                                                             |
|---------------------------------------------------------------------------------------------|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| rules/src/<br>lib.rs:1090                                                                   |                            | 1092       /// Mahdi's Tomb control at conclusion of play<br>(\$9.14).                                                                                                                                                                                                                                                      |
| omdurman-<br>rules/src/<br>lib.rs:1116<br>GH:omdurman-<br>rules/src/<br>lib.rs:1116         | VpSource::points           | 1114   impl VpSource {<br>1115       /// VP awarded to `who_scores()` (rulebook \$9.14).<br>1116       pub fn points(self) -> VictoryPoints {<br>1117           match self {<br>1118               VpSource::MahdisTomb => VictoryPoints(25),                                                                               |
| omdurman-<br>rules/src/<br>lib.rs:1131<br>GH:omdurman-<br>rules/src/<br>lib.rs:1131         | VpSource::who_scores       | 1129  <br>1130       /// Which player receives these victory points<br>(rulebook \$9.14).<br>1131       pub fn who_scores(self) -> Player {<br>1132           match self {<br>1133               VpSource::MahdisTomb                                                                                                       |
| omdurman-<br>rules/src/<br>lib.rs:1148<br>GH:omdurman-<br>rules/src/<br>lib.rs:1148         | VictoryLedger              | 1146       /// Cumulative victory ledger for one scenario<br>(rulebook \$9.14).<br>1147       #[derive(Serialize, Deserialize, Clone, Debug,<br>Default)]<br>1148       pub struct VictoryLedger {<br>1149           pub events: Vec<VpEvent>,<br>1150       }                                                              |
| omdurman-<br>rules/src/<br>lib.rs:1154<br>GH:omdurman-<br>rules/src/<br>lib.rs:1154         | VpEvent                    | 1152       /// A single victory-point scoring event (rulebook<br>\$9.14).<br>1153       #[derive(Serialize, Deserialize, Clone, Copy,<br>PartialEq, Eq, Debug)]<br>1154       pub struct VpEvent {<br>1155           pub turn: GameTurnIndex,<br>1156           pub source: VpSource,                                       |
| omdurman-<br>rules/src/<br>lib.rs:1161<br>GH:omdurman-<br>rules/src/<br>lib.rs:1161         | VictoryLedger::total_for   | 1159   impl VictoryLedger {<br>1160       /// Total victory points earned by a given player<br>(rulebook \$9.14).<br>1161       pub fn total_for(&self, player: Player) -><br>VictoryPoints {<br>1162           VictoryPoints(<br>1163               self.events                                                            |
| omdurman-<br>rules/src/<br>lib.rs:1173<br>GH:omdurman-<br>rules/src/<br>lib.rs:1173         | VictoryLedger::superiority | 1171       /// Net superiority: positive = Anglo-Egyptian<br>ahead, negative = Dervish ahead<br>1172       /// (rulebook \$9.14).<br>1173       pub fn superiority(&self) -> VictoryPoints {<br>1174       VictoryPoints(self.total_for(Player::AngloEgyptian).0 -<br>self.total_for(Player::Dervish).0)<br>1175       }    |
| omdurman-<br>rules/src/<br>lib.rs:1191<br>GH:omdurman-<br>rules/src/<br>lib.rs:1191         | CampaignVictoryLevel       | 1189       /// Campaign-game victory levels (\$9.14).<br>1190       #[derive(Serialize, Deserialize, Clone, Copy,<br>PartialEq, Eq, Debug)]<br>1191       pub enum CampaignVictoryLevel {<br>1192           Draw,<br>1193           Marginal(Player),                                                                       |
| omdurman-<br>rules/src/<br>lib.rs:1200<br>GH:omdurman-<br>rules/src/<br>lib.rs:1200         | CampaignVictoryLevel::from | 1198   impl CampaignVictoryLevel {<br>1199       /// Assign a level from the net superiority<br>(\$9.14).<br>1200       pub fn from_superiority(s: VictoryPoints) -> Self<br>{<br>1201           let net = s.0;<br>1202           // Positive -> Anglo-Egyptian thresholds:<br>15/30/50                                     |
| omdurman-<br>rules/src/<br>effects.rs:3195<br>GH:omdurman-<br>rules/src/<br>effects.rs:3195 | score_elimination          | 3193  <br>3194       /// Score victory points for eliminating a unit<br>(rulebook \$9.14).<br>3195       pub fn score_elimination(state: &mut GameState,<br>unit_id: UnitId, _owner: Player) {<br>3196           if let Some(unit) = state.find_unit(unit_id) {<br>3197               let identity = unit.profile.identity; |

|                                                                                   |               |                                                                                                                                                                                                                                                                                     |
|-----------------------------------------------------------------------------------|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/lib.rs:227</a><br>GH:omdurman-rules/src/lib.rs:227 | VictoryPoints | <pre>225   /// (rulebook §9.14). 226   #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, PartialOrd, Ord, Debug, Default)] 227   pub struct VictoryPoints(pub i32); 228   229   /// One-based Game Turn index (1, 2, ... up to the scenario length) (rulebook §4).</pre> |
|-----------------------------------------------------------------------------------|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Covered by tests: [friendlies\\_bank\\_scores\\_by\\_side](#) [mahdis\\_tomb\\_not\\_scored\\_without\\_a\\_leader](#) [mahdis\\_tomb\\_scores\\_for\\_anglo\\_egyptian\\_when\\_held](#) [vp\\_source\\_attributes](#)

§9.21 -- Set Up (Historical)

[out-of-scope](#)  
manual page unknown

Set Up

§9.22 -- Scenario Length (Historical)

[implemented](#)  
manual page unknown

Scenario Length

6:00 am, September 2 through 12:00 noon, September 2. Four game turns.

| File                                                                                            | Symbol                | Code Snippet                                                                                                                                                                                                   |
|-------------------------------------------------------------------------------------------------|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/turn_track.rs:129</a><br>GH:omdurman-rules/src/turn_track.rs:129 | HISTORICAL_TURN_TRACK | <pre>127   128   /// Historical scenario track (§9.22 -- 4 turns, Sept 2 6:00 am -&gt; 12:00 pm). 129   pub const HISTORICAL_TURN_TRACK: [TurnEntry; 4] = [ 130       TurnEntry { 131           turn: 1,</pre> |

Covered by tests: [historical\\_turn\\_all\\_four\\_turns](#)

§9.23 -- Special Rule: The Zariba

[implemented](#)  
manual page unknown

Special Rule: “The Zariba”

**\*\*9.231)** Thorn hedge hexsides:**\*\*** −2 to die roll on all Dervish fire attacks; may not melee across in either direction; may not advance after combat across in either direction.

**\*\*9.232)** Trench hexsides:**\*\*** −4 to die roll on all Dervish fire attacks vs. entrenched units only; −2 (instead of +2) melee modifier to Dervish units melee attacking an entrenched unit; entrenched units may be fired “over” in both directions (i.e. they do not block line of sight); units are considered “entrenched” if they are directly adjacent to (and on the Nile River side of) a trench hexside.

(see manual for full text)

See also: §9.231, §9.232

| File | Symbol | Code Snippet |
|------|--------|--------------|
|------|--------|--------------|

|                                                                       |                        |                                                                                                                                                                                                                                                                                                      |
|-----------------------------------------------------------------------|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| omdurman-rules/src/<br>lib.rs:873<br>GH:omdurman-rules/src/lib.rs:873 | ZaribaThornHedge       | 871   Terrain(i16),<br>872   /// -2 thorn-hedge defensive modifier (§9.231).<br>873   ZaribaThornHedge,<br>874   /// -4 trench defensive modifier (§9.232). Only<br>applies vs. "entrenched"<br>875   /// units (those Nile-side of the trench hexside).                                             |
| omdurman-rules/src/<br>lib.rs:876<br>GH:omdurman-rules/src/lib.rs:876 | ZaribaTrenchEntrenched | 874   /// -4 trench defensive modifier (§9.232). Only<br>applies vs. "entrenched"<br>875   /// units (those Nile-side of the trench hexside).<br>876   ZaribaTrenchEntrenched,<br>877   }<br>878                                                                                                     |
| omdurman-types/src/<br>lib.rs:212<br>GH:omdurman-types/src/lib.rs:212 | ZaribaThornHedge       | 210   Crest,<br>211   /// Historical-scenario thorn-hedge segment of the<br>Zariba (§9.231).<br>212   ZaribaThornHedge,<br>213   /// Historical-scenario trench segment of the<br>Zariba (§9.232).<br>214   ZaribaTrench,                                                                            |
| omdurman-types/src/<br>lib.rs:214<br>GH:omdurman-types/src/lib.rs:214 | ZaribaTrench           | 212   ZaribaThornHedge,<br>213   /// Historical-scenario trench segment of the<br>Zariba (§9.232).<br>214   ZaribaTrench,<br>215   /// One of the two end hexsides of a Zariba trench<br>segment that connect to<br>216   /// the Nile River (§9.233). Units may only enter/<br>leave the Zariba via |

## §9.24 -- Victory Conditions (Historical)

implemented

manual page unknown

### Victory Conditions

Victory Levels are based solely on eliminating enemy units while conserving your own force as much as possible.

| Victory Level | Anglo-Egyptian Player (Dervish units eliminated) | Dervish Player (Anglo-Egyptian units eliminated)  
| |—|—|—| | 5 — DECISIVE | 100+ | 30+ | | 4 — STRATEGIC | 60–99 | 15–29 | | 3 — TACTICAL | 45–59 | 10–14 | | 2 — MARGINAL | 30–44 | 5–9 | | 1 — DRAW | 0–29 | 0–4 |

The lower value victory level is then subtracted from the higher level to determine a player's net victory. For example, if the Anglo-Egyptian player eliminates 104 Dervish units (decisive victory) but loses 18 units doing it (Dervish Strategic), the Anglo-Egyptian player only nets out with a draw (decisive worth 5 minus strategic worth 4 = 1, draw).

(see manual for full text)

| File                                                                        | Symbol                 | Code Snippet                                                                                                                                                                                                          |
|-----------------------------------------------------------------------------|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| omdurman-rules/src/<br>lib.rs:1230<br>GH:omdurman-rules/src/<br>lib.rs:1230 | HistoricalVictoryLevel | 1228   /// draw").<br>1229   #[derive(Serialize, Deserialize, Clone, Copy,<br>PartialEq, Eq, PartialOrd, Ord, Debug)]<br>1230   pub enum HistoricalVictoryLevel {<br>1231       Draw = 1,<br>1232       Marginal = 2, |

## §9.31 -- Bonus game map

out-of-scope

manual page unknown

Only the small FALL OF KHARTOUM scenario map is used for this game.



§9.32 -- Set Up (Bonus)

out-of-scope  
manual page unknown

Set Up

§9.33 -- Scenario Length (Bonus)

implemented  
manual page unknown

Scenario Length

Variable, see victory conditions (9.35). Rarely lasts five turns.

See also: §9.35

| File                                                                                            | Symbol           | Code Snippet                                                                                                                                                                                                                                       |
|-------------------------------------------------------------------------------------------------|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/turn_track.rs:169</a><br>GH:omdurman-rules/src/turn_track.rs:169 | FALL_OF_KHARTOUM | <pre>167   /// (the rulebook fixes none); only `day_night` is 168   /// Anglo-Egyptian movement and ranges and bars 169   pub const FALL_OF_KHARTOUM_TURN_TRACK: [TurnEntry; 8] = 170       [ 171           TurnEntry {             turn: 1,</pre> |

Covered by tests: [fall\\_of\\_khartoum\\_turn\\_one\\_is\\_night](#) [fall\\_of\\_khartoum\\_turns\\_3\\_to\\_8\\_are\\_day](#)

§9.34 -- Special Rules (Bonus)

out-of-scope  
manual page unknown

Special Rules

§9.35 -- Victory Conditions (Bonus)

implemented  
manual page unknown

Victory Conditions

Victory is determined by how many turns it takes the Dervish player to eliminate the GORDON leader unit and how many Dervish units are eliminated:

- Dervish decisive: eliminate GORDON turn four or sooner.
- Dervish tactical: eliminate GORDON turn five.
- Dervish marginal: eliminate GORDON turn six.
- British marginal: GORDON survives end of turn six.
- British tactical: GORDON survives end of turn seven.
- British decisive: GORDON survives end of turn eight.

(see manual for full text)

| File                                           | Symbol          | Code Snippet                                                                                                               |
|------------------------------------------------|-----------------|----------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/lib.rs:1271</a> | FoKVictoryLevel | <pre>1269   /// negative) so the loss penalty is a simple shift 1270   #[derive(Serialize, Deserialize, Clone, Copy,</pre> |

|                                   |  |                                                                                                                                                        |
|-----------------------------------|--|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| GH:omdurman-rules/src/lib.rs:1271 |  | <pre>PartialEq, Eq, PartialOrd, Ord, Debug]] 1271   pub enum FoKVictoryLevel { 1272       DervishDecisive = -3, 1273       DervishTactical = -2,</pre> |
|-----------------------------------|--|--------------------------------------------------------------------------------------------------------------------------------------------------------|

§9.111 -- Dervish set up (Campaign)

out-of-scope  
manual page unknown

Dervish player sets up first, moves second.

- Isa Zachneih infantry unit: anywhere on the east bank, in or south of El Debeba.
- KHALIFA ABDULLAH: in the walled city of Omdurman, in either palace hex.
- 3 artillery units, and all Taiasha units: anywhere in the walled city of Omdurman.
- 17 forts: anywhere on the mapsheet south of the Khor Shambat on the west bank, and/or south of all Halfaya hut hexes on the east bank and Nile River islands.
- 2 gunboats: any south edge Nile River hexes.

**\*\*9.112) Dervish reinforcements:\*\*** all reinforcements enter on the west edge of the mapsheet, south of the Khor Shambat. Each unit pays the terrain cost of the hex through which it enters, no matter how many units enter through  
(see manual for full text)

See also: §9.112

§9.112 -- Dervish reinforcements (Campaign)

implemented  
manual page unknown

| File                                                                        | Symbol                    | Code Snippet                                                                                                                                                                                                                                                       |
|-----------------------------------------------------------------------------|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| omdurman-rules/src/effects.rs:120<br>GH:omdurman-rules/src/effects.rs:120   | PlaceReinforcements       | <pre>118   // -- Reinforcement / placement ----- 119   /// Place reinforcements onto the map (rulebook \$9.112, \$9.113). 120       PlaceReinforcements(Vec&lt;UnitPlacement&gt;), 121   122   // -- Scenario-specific -----</pre>                                 |
| omdurman-rules/src/effects.rs:2678<br>GH:omdurman-rules/src/effects.rs:2678 | apply_place_reinforcement | <pre>2676   2677   /// Place reinforcements onto the map (rulebook \$9.112, \$9.113). 2678   pub fn apply_place_reinforcements( 2679       state: &amp;mut GameState, 2680       placements: &amp;[UnitPlacement],</pre>                                           |
| omdurman-types/src/lib.rs:473<br>GH:omdurman-types/src/lib.rs:473           | Location                  | <pre>471   /// Named map landmarks (rulebook mapsheet, \$9.111, \$9.113, \$9.212 scenarios). 472   #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug, strum::Display)] 473   pub enum Location { 474       FortMakran, 475       NorthFort,</pre> |
| omdurman-types/src/lib.rs:525<br>GH:omdurman-types/src/lib.rs:525           | SetupLetter               | <pre>523   /// Each letter marks a specific hex where a Dervish leader is placed. 524   #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug, strum::Display)] 525   pub enum SetupLetter { 526       Y, 527       K,</pre>                          |
| omdurman-types/src/                                                         | Faction                   | <pre>617   /// Dervish units have a tribe; Anglo-Egyptian infantry have a brigade.</pre>                                                                                                                                                                           |

|                                                                                |  |                                                                                                                                                                                                             |
|--------------------------------------------------------------------------------|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">lib.rs:619</a><br><a href="#">GH:omdurman-types/src/lib.rs:619</a> |  | <pre> 618   #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 619   pub enum Faction { 620       Dervish { tribe: DervishTribe }, 621       BritishEgyptian { brigade: Brigade }, </pre> |
|--------------------------------------------------------------------------------|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## §9.113 -- Anglo-Egyptian set up (Campaign)

out-of-scope

manual page unknown

The Anglo-Egyptian player moves first. There are no Anglo-Egyptian units on the mapsheet at start. The GORDON unit is not used in this scenario.

- The leader units KITCHENER, GATACRE, and HUNTER may enter anytime during the first four game turns and do not count against the 12 unit per turn limit. All three leaders must be in play by the end of turn four!
- All gunboats enter through any north edge Nile River hex, paying one movement point for the first hex entered. The “Friendlies” brigade enters through the Abu Alim hut hex on the east bank, paying eight movement points per unit. All other Anglo-Egyptian units enter through the west bank “ANGLO-EGYPTIAN ENTRANCE AREA”, each unit paying one movement point to enter the mapsheet.
- Turn 1) Any three gunboats; “Friendlies” brigade; Egyptian Cavalry; Horse Artillery; and two infantry brigades

(see manual for full text)

## §9.211 -- Anglo-Egyptian set up first, moves second (Historical)

implemented

manual page unknown

The Anglo-Egyptian player sets up first, and moves second:

- Not in play: GORDON leader unit, “Friendlies” brigade.
- Gunboats start in any Nile River hexes adjacent to the Zariba.
- Camel Corps, Egyptian Cavalry, and Horse Artillery start in the village of Kerreri hut hexes.
- All remaining Anglo-Egyptian units set up in the 13 hexes of the Zariba.

| File                                                                                                        | Symbol                    | Code Snippet                                                                                                                                                                                                                          |
|-------------------------------------------------------------------------------------------------------------|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/effects.rs:1629</a><br><a href="#">GH:omdurman-rules/src/effects.rs:1629</a> | <code>first_player</code> | <pre> 1627   1628   /// The player who moves first in a scenario (§4, 1629   pub fn first_player(scenario: Scenario) -&gt; Player { 1630       match scenario { 1631           Scenario::Campaign =&gt; Player::AngloEgyptian, </pre> |

## §9.212 -- Dervish set up (Historical) -- deployment zones

implemented

manual page unknown

The Dervish player sets up second, and moves first.

- Not in play: Isa Zachneih, gunboats, and forts.
- All Dervish units must be set up out of the line of sight of all Anglo-Egyptian units.
- Dervish leaders start on the lettered hexes:
  - A: Ali Wad Helu
  - D: Sheik El Din
  - Y: Yakub

- K: Khalifa Abdullah
- S: Sherif

(see manual for full text)

| File                                                                                      | Symbol                    | Code Snippet                                                                                                                                                                                                                                                                                                                                                        |
|-------------------------------------------------------------------------------------------|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/effects.rs:629</a><br>GH:omdurman-rules/src/effects.rs:629 | <b>in_deployment_zone</b> | <pre> 627       /// plan / UI rather than this hex predicate.         Documented, not silently 628       /// dropped. 629       pub fn in_deployment_zone(&amp;self, player: Player,         hex: HexCoord) -&gt; bool { 630           // No board attached -&gt; permissive (unit tests,         sandbox). 631           if self.board.terrain.is_empty() { </pre> |

## §9.231 -- Thorn hedge hexsides

implemented

manual page unknown

| File                                                                              | Symbol                  | Code Snippet                                                                                                                                                                                                                                                                       |
|-----------------------------------------------------------------------------------|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/lib.rs:873</a><br>GH:omdurman-rules/src/lib.rs:873 | <b>ZaribaThornHedge</b> | <pre> 871       Terrain(i16), 872       /// -2 thorn-hedge defensive modifier (§9.231). 873       ZaribaThornHedge, 874       /// -4 trench defensive modifier (§9.232). Only         applies vs. "entrenched" 875       /// units (those Nile-side of the trench hexside). </pre> |
| <a href="#">omdurman-types/src/lib.rs:212</a><br>GH:omdurman-types/src/lib.rs:212 | <b>ZaribaThornHedge</b> | <pre> 210       Crest, 211       /// Historical-scenario thorn-hedge segment of the         Zariba (§9.231). 212       ZaribaThornHedge, 213       /// Historical-scenario trench segment of the         Zariba (§9.232). 214       ZaribaTrench, </pre>                           |

## §9.232 -- Trench hexsides

implemented

manual page unknown

| File                                                                              | Symbol                        | Code Snippet                                                                                                                                                                                                                                                                                                                             |
|-----------------------------------------------------------------------------------|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/lib.rs:876</a><br>GH:omdurman-rules/src/lib.rs:876 | <b>ZaribaTrenchEntrenched</b> | <pre> 874       /// -4 trench defensive modifier (§9.232). Only         applies vs. "entrenched" 875       /// units (those Nile-side of the trench hexside). 876       ZaribaTrenchEntrenched, 877       } 878   </pre>                                                                                                                 |
| <a href="#">omdurman-types/src/lib.rs:214</a><br>GH:omdurman-types/src/lib.rs:214 | <b>ZaribaTrench</b>           | <pre> 212       ZaribaThornHedge, 213       /// Historical-scenario trench segment of the         Zariba (§9.232). 214       ZaribaTrench, 215       /// One of the two end hexsides of a Zariba trench         segment that connect to 216       /// the Nile River (§9.233). Units may only enter/         leave the Zariba via </pre> |

## §9.233 -- Zariba entry/exit costs

implicit

manual page unknown

Units may only enter and/or leave the Zariba via the two end hexsides that connect to the Nile River, paying +2 movement points to cross (Exception: advance after combat across an entrenched hexside).

§9.321 -- British set up (Bonus)

out-of-scope  
manual page unknown

The British player sets up first, moves second:

- General GORDON leader unit in the palace.
- Two old style (unnamed) gunboats in any Nile River hexes.
- Set up in any building or hut hexes of Khartoum, Forts Makran and/or Buri, and/or adjacent to any wall hex:
  - one Egyptian Battalion artillery unit
  - two British infantry units (represents Caucasian troops)
  - three Egyptian infantry units (represents Cairo “Bazouks”)
  - four Sudan infantry units (represents Sudanese blacks)
  - four “Friendlies” units (represents the Shaggyeh)

(see manual for full text)

§9.322 -- Dervish enters turn one (Bonus)

out-of-scope  
manual page unknown

Dervish player moves first: enters turn one through any hexes on the south or east edge of the map.

- 32 Mulazmin units (represents combined forces of Wad El Nejumi, Abu Girgeh, and Sheik El Obeid)
- 2 Hadendowa; 6 Kehena; 5 Degheim (represents Mahdi’s combined west bank forces)
- 3 Dervish artillery units.

§9.341 -- Turn 1 is always a night turn (Bonus)

implemented  
manual page unknown

Turn 1 is always a night turn (see 8.1).

See also: §8.1

| File                                                                            | Symbol           | Code Snippet                                                                                                                                                                                                                                                                                      |
|---------------------------------------------------------------------------------|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| omdurman-rules/src/turn_track.rs:169<br>GH:omdurman-rules/src/turn_track.rs:169 | FALL_OF_KHARTOUM | 167   /// (the rulebook fixes none); only `day_night` is<br>rule-bearing (night halves<br>168   /// Anglo-Egyptian movement and ranges and bars<br>howitzer fire, §8.1).<br>169   pub const FALL_OF_KHARTOUM_TURN_TRACK: [TurnEntry; 8] =<br>[<br>170       TurnEntry {<br>171           turn: 1, |

Covered by tests: fall\_of\_khartoum\_turn\_one\_is\_night

§9.343 -- Both players use the Dervish Range Effects Table (Bonus)

implemented  
manual page unknown

Both players must use the Dervish Range Effects Table.

| File                                                                                        | Symbol                | Code Snippet                                                                                                                                                                                                     |
|---------------------------------------------------------------------------------------------|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/effects.rs:1957</a><br>GH:omdurman-rules/src/effects.rs:1957 | <b>range_band_for</b> | <pre>1955   /// in FALL OF KHARTOUM *both* players use the Dervish 1956   Range Effects Table 1957   /// (§9.343). 1958   pub fn range_band_for( 1959       scenario: Scenario, 1959       player: Player,</pre> |

### §9.344 -- Dervish controls the North Fort (Bonus)

implemented

manual page unknown

The Dervish player controls the “North Fort” and may fire its guns.

| File                                                                                        | Symbol                    | Code Snippet                                                                                                                                                                                                                                                                                                                                                                               |
|---------------------------------------------------------------------------------------------|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/effects.rs:1182</a><br>GH:omdurman-rules/src/effects.rs:1182 | <b>hex_has_enemy_fort</b> | <pre>1180       /// may neither occupy an enemy fort nor advance 1181       after combat into one 1182       /// (forts are never captured -- only destroyed, 1183       \$6.62/\$6.53/\$7.6). 1184       pub fn hex_has_enemy_fort(&amp;self, hex: HexCoord, 1185       mover: Player) -&gt; bool { 1186           self.units.iter().any( u  { 1187               u.position == hex</pre> |

### §9.345 -- Gunboat White Nile <-> Blue Nile crossing (Bonus)

implemented

manual page unknown

The British gunboats may move from the White Nile to the Blue Nile and vice-versa at an off-board movement cost of six “upstream” movement points.

| File                                                                                        | Symbol                           | Code Snippet                                                                                                                                                                                                                                                                                                                           |
|---------------------------------------------------------------------------------------------|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/effects.rs:1166</a><br>GH:omdurman-rules/src/effects.rs:1166 | <b>is_nile_mouth_crossing</b>    | <pre>1164       /// must be named on the board, else this is 1165       `false` and the move falls 1166       /// through to the ordinary contiguous-Nile rules. 1167       pub fn is_nile_mouth_crossing(&amp;self, from: 1168       HexCoord, to: HexCoord) -&gt; bool { 1169           let white = self 1170           .board</pre> |
| <a href="#">omdurman-types/src/lib.rs:489</a><br>GH:omdurman-types/src/lib.rs:489           | <b>Location::WhiteNile-Mouth</b> | <pre>487       /// The off-board mouth of the White Nile branch 488       (FALL OF KHARTOUM §9.345) -- 489       /// a British gunboat may cross to the Blue Nile 490       mouth for 6 upstream MP. 491       WhiteNileMouth,</pre>                                                                                                   |

### §9.346 -- GORDON immobile, eliminated only at the Palace (Bonus)

implemented

manual page unknown

The GORDON leader unit starts in the palace and may not move during the scenario. He may only be eliminated by a Dervish unit passing through or occupying the palace hex (as normal movement or as advance after combat).

| File                                                                                        | Symbol                               | Code Snippet                                                                                                                                                                                                                                                                                                                               |
|---------------------------------------------------------------------------------------------|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/effects.rs:1769</a><br>GH:omdurman-rules/src/effects.rs:1769 | <code>check_gordon_palace</code>     | <pre>1767   /// after combat). Records the turn (which fixes the 1768   /// ends the game. A no-op outside FoK, or once GORDON 1769   pub fn check_gordon_palace(state: &amp;mut GameState) { 1770       if state.scenario != Scenario::FallOfKhartoum    1771       state.gordon_eliminated_turn.is_some() { 1772           return;</pre> |
| <a href="#">omdurman-rules/src/lib.rs:651</a><br>GH:omdurman-rules/src/lib.rs:651           | <code>UnitIdentity::is_gordon</code> | <pre>649   /// Whether this is the GORDON leader unit (§9.32, 650   §9.346) -- the immobile 651   /// palace defender whose elimination ends FALL OF 652   KHARTOUM (§9.35). 653   pub fn is_gordon(&amp;self) -&gt; bool { 654       matches!( 655           self,</pre>                                                                  |

Covered by tests: `gordon_survives_means_no_elimination`

 7/10 implemented

## §10 -- Optional Rules

### §10 -- Optional Rules

implemented

manual page unknown

#### Optional Rules (Campaign game only)

It is suggested that the most intriguing employment of the following two options is to permit the Dervish player to have either one or the other, but the Anglo-Egyptian player doesn't know which one until he stumbles onto it. Players are advised that the employment of both optionals in the same game is not recommended.

| File                                                                              | Symbol       | Code Snippet                                                                                                                                                                                                    |
|-----------------------------------------------------------------------------------|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/lib.rs:326</a><br>GH:omdurman-rules/src/lib.rs:326 | OptionalRule | <pre>324   /// two should be in play (rulebook §10). 325   #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 326   pub enum OptionalRule { 327       RiverMines, 328       RiverChain,</pre> |

### §10.1 -- River Mines

implemented

manual page unknown

#### River Mines

The Khalifa twice tried (unsuccessfully) to submerge a powerful mine in the Nile to sink or damage British gunboats. This option assumes that both attempts were successful.

| File                                                                                        | Symbol           | Code Snippet                                                                                                                                                                                                                |
|---------------------------------------------------------------------------------------------|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/effects.rs:138</a><br>GH:omdurman-rules/src/effects.rs:138   | RiverMine        | <pre>136   // -- Optional rules ----- 137   /// River mine resolution (rulebook §10.12). 138   RiverMine { 139       gunboat_id: UnitId, 140       hex: HexCoord,</pre>                                                     |
| <a href="#">omdurman-rules/src/effects.rs:2928</a><br>GH:omdurman-rules/src/effects.rs:2928 | apply_river_mine | <pre>2926   2927   /// Apply a river-mine resolution (rulebook §10.12). 2928   pub fn apply_river_mine( 2929       state: &amp;mut GameState, 2930       gunboat_id: UnitId,</pre>                                          |
| <a href="#">omdurman-rules/src/lib.rs:1042</a><br>GH:omdurman-rules/src/lib.rs:1042         | MineResult       | <pre>1040   /// British gunboat enters a mined hex. 1041   #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 1042   pub enum MineResult { 1043       /// Roll 1-4: no effect. 1044       NoEffect,</pre> |

### §10.2 -- River Chain

out-of-scope

manual page unknown

#### River Chain



The Khalifa also tried (also unsuccessfully) to string a heavy chain across the Nile to stop or slow down the British gunboats. This option assumes the chain was emplaced.

## §10.11 -- Secretly record mine hexes

out-of-scope

manual page unknown

Prior to the commencement of play the Dervish player secretly records two Nile River hexes to be mined (the mines may not both be placed in the same hex). These hexes must be south of the E–W hexrow in which the Khor Shambat empties into the Nile.

Covered by tests: [mine\\_and\\_chain\\_limits\\_enforced\\_in\\_setup](#)

## §10.12 -- Mine resolution

implemented

manual page unknown

When a British gunboat enters a mined hex, the Dervish player must order it to stop as it has struck a mine. The Dervish player then resolves the effect of the mine's blast by rolling the ten-sided die:

- 1–4: No effect
- 5–7: Gunboat damaged, lost use of its engines and must drift two hexes per turn (with the current) for the rest of the game. No effect on guns or Maxims unless they drift out of range.
- 8–10: Gunboat sunk!

| File                                                                                        | Symbol                  | Code Snippet                                                                                                                                                                       |
|---------------------------------------------------------------------------------------------|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/effects.rs:138</a><br>GH:omdurman-rules/src/effects.rs:138   | <b>RiverMine</b>        | <pre>136   // -- Optional rules       ----- 137   /// River mine resolution (rulebook §10.12). 138   RiverMine { 139       gunboat_id: UnitId, 140       hex: HexCoord,</pre>      |
| <a href="#">omdurman-rules/src/effects.rs:2928</a><br>GH:omdurman-rules/src/effects.rs:2928 | <b>apply_river_mine</b> | <pre>2926   2927   /// Apply a river-mine resolution (rulebook §10.12). 2928   pub fn apply_river_mine( 2929       state: &amp;mut GameState, 2930       gunboat_id: UnitId,</pre> |

## §10.13 -- Mines consumed after both rolled for

implemented

manual page unknown

After both mines have been rolled for, no more are available.

| File                                                                                        | Symbol                  | Code Snippet                                                                                                                                                                       |
|---------------------------------------------------------------------------------------------|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/effects.rs:2928</a><br>GH:omdurman-rules/src/effects.rs:2928 | <b>apply_river_mine</b> | <pre>2926   2927   /// Apply a river-mine resolution (rulebook §10.12). 2928   pub fn apply_river_mine( 2929       state: &amp;mut GameState, 2930       gunboat_id: UnitId,</pre> |

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The Dervish player's gunboats may pass through the mined hexes with no ill effect (he knows where the mines are).

| File                                                                                                        | Symbol           | Code Snippet                                                                                                                                                                         |
|-------------------------------------------------------------------------------------------------------------|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/effects.rs:2928</a><br><a href="#">GH:omdurman-rules/src/effects.rs:2928</a> | apply_river_mine | <pre> 2926   2927   /// Apply a river-mine resolution (rulebook §10.12). 2928   pub fn apply_river_mine( 2929       state: &amp;mut GameState, 2930       gunboat_id: UnitId, </pre> |

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Prior to the commencement of play the Dervish player secretly records a line of river hexes (not exceeding four hexes long) across which the chain is strung. The hexes must be south of the E-W hexrow in which the Khor Shambat empties into the Nile.

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When a British gunboat enters a “chained” river hex it must stop and may move no further that turn.

| File                                                                                      | Symbol                        | Code Snippet                                                                                                                                                                                                                                                                                            |
|-------------------------------------------------------------------------------------------|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/effects.rs:872</a><br>GH:omdurman-rules/src/effects.rs:872 | <code>can_move_gunboat</code> | <pre> 870           /// upstream movement allowance is their maximum                 for that turn." Chained Nile 871           /// hexes stop the gunboat (\$10.22).         872           pub fn can_move_gunboat(                     &amp;self, 873               unit_id: UnitId,           </pre> |

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No gunboats (British or Dervish) may cross the chain until it has been sunk by the British player. He may sink the chain by a) having an infantry or cavalry unit spend one complete turn on either riverbank adjacent to a “chained” river hex, or b) firing at the chain with artillery and achieving a 3 or more on the Combat Results Table.

| File                                                                                                        | Symbol                        | Code Snippet                                                                                                                                                                                                                                                                                                                                               |
|-------------------------------------------------------------------------------------------------------------|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/effects.rs:2987</a><br><a href="#">GH:omdurman-rules/src/effects.rs:2987</a> | <code>apply_sink_chain</code> | <pre> 2985   /// Sink the river chain (rulebook §10.23). Marks the         placed chain cleared so it 2986   /// no longer stops gunboats (§10.22). 2987   pub fn apply_sink_chain(state: &amp;mut GameState) -&gt;         Result&lt;(), RuleError&gt; { 2988       match state.chain.as_mut() { 2989           Some(chain) if !chain.sunk =&gt; { </pre> |

## §11 -- Historical Notes

### §11 -- Historical Notes

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manual page unknown

#### Historical Notes

In 1881 Mohammed Ahmed Ibn Al-Sayid Abdullah, the son of an obscure carpenter in the hinterlands of the Sudan, proclaimed himself the “Mahdi” — the Messiah of the Islamic faith. His timing was propitious indeed. Since the early 1820’s a corrupt Egypt, with the Sultan of Turkey’s blessing, had incessantly raped the Sudan, taking ivory and flooding the slave markets with some half million captured Sudanese blacks. By 1880, nearly 40,000 Egyptian troops occupied outposts scattered throughout the Sudan, enforcing Egypt’s hold on this lucrative ivory and slave trade and squeezing the native population dry through vicious and corrupt tax officials. All was controlled from Khartoum via the office of Governor General of the Sudan. The title had been held by a succession of individuals, including General Charles Gordon, whose appointment was an attempt to reinstate some rudimentary justice in the Sudan after France and Britain assumed joint political control of a bankrupt Egypt.

By 1881, however, Gordon’s term had expired and a new Governor General, again corrupt and incompetent,  
(see manual for full text)

See also: §9.1, §9.2, §9.3

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## Credits

§Credits -- Credits

[descriptive](#)

manual page unknown

 1/1 implemented

## Combat Results Table (shared reference)

§CRT -- Combat Results Table (shared by §6.22 fire and §7.7 melee)

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manual page unknown

| File                                                                                                              | Symbol                      | Code Snippet                                                                                                                                                                                                                                                                |
|-------------------------------------------------------------------------------------------------------------------|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/combat_results_table.rs:8</a><br>GH:omdurman-rules/src/combat_results_table.rs:8   | <b>FireFactorRow</b>        | <pre>6   /// to index into the result matrix. 7   #[derive(serde::Serialize, serde::Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 8   pub enum FireFactorRow { 9       /// 1-5 factors 10       Row01to05,</pre>                                                         |
| <a href="#">omdurman-rules/src/combat_results_table.rs:31</a><br>GH:omdurman-rules/src/combat_results_table.rs:31 | <b>from_total</b>           | <pre>29   impl FireFactorRow { 30       /// Determine which row a given total fire factor falls into (rulebook §6.22). 31       pub fn from_total(total: u16) -&gt; Self { 32           match total { 33               0..=5 =&gt; FireFactorRow::Row01to05,</pre>          |
| <a href="#">omdurman-rules/src/combat_results_table.rs:69</a><br>GH:omdurman-rules/src/combat_results_table.rs:69 | <b>combat_results_table</b> | <pre>67   /// D = `Disrupt` (1/2 of target units, round up) 68   /// 1...5 = `Eliminate(n)` (that many units removed) 69   pub fn combat_results_table(row: FireFactorRow, roll: DieRoll) -&gt; CombatResult { 70       use CombatResult::*; 71       use DieRoll::*;</pre> |
| <a href="#">omdurman-rules/src/lib.rs:937</a><br>GH:omdurman-rules/src/lib.rs:937                                 | <b>CombatResult</b>         | <pre>935   /// * `--` -- no effect 936   #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 937   pub enum CombatResult { 938       NoEffect, 939       Disrupt,</pre>                                                                                    |

Covered by tests: [ae\\_combat\\_results\\_table\\_lowest\\_is\\_no\\_effect](#)  
[ae\\_combat\\_results\\_table\\_highest\\_is\\_eliminate\\_5](#) [ae\\_combat\\_results\\_table\\_progresses\\_with\\_roll](#) [ae\\_combat\\_results\\_table\\_progresses\\_with\\_factor](#)  
[fire\\_factor\\_row\\_boundaries](#) [fire\\_factor\\_row\\_remaining\\_boundaries](#) [fire\\_factor\\_row\\_index\\_sequential](#) [crt\\_all\\_rows\\_monotone\\_non\\_decreasing](#)  
[crt\\_eliminate\\_never\\_exceeds\\_5](#)

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## Reference -- Charts and Tables

\$Reference -- Charts and Tables

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manual page unknown

## Appendix: Symbol Index

| Symbol                       | Sections      |
|------------------------------|---------------|
| AdvanceAfterCombat           | §6.82, §7.6   |
| AdvancePhase                 | §4            |
| AngloEgyptianDirectFire      | §6.24         |
| AngloEgyptianStandard        | §7.7          |
| BattalionOrdinal             | §5.54         |
| Breach                       | §6.63         |
| Brigade                      | §5.54         |
| BrigadeId                    | §2.3          |
| BrigadeIntegrity             | §5.54, §6.24  |
| BritishLeader                | §6.51         |
| BritishLeaderUnit            | §6.51         |
| CAMPAIGN_TURN_TRACK          | §9.12         |
| CampaignVictoryLevel         | §9.14         |
| CombatResult                 | §CRT          |
| ConstructZariba              | §5.3          |
| DayNight                     | §8.1          |
| Demolition                   | §6.53         |
| DemolitionTarget             | §6.53         |
| DervishDesertion             | §8.2          |
| DervishGunboat               | §2.32         |
| DervishLeaderCommandMismatch | §5.53         |
| DervishStandard              | §7.7          |
| DervishTribe                 | §7.4          |
| DervishTribeMix              | §5.52         |
| DervishVsTrenchedDefender    | §7.7          |
| DieModifier                  | §7.7          |
| DirectFire                   | §6.41         |
| FALL_OF_KHARTOUM_TURN_TRACK  | §9.33, §9.341 |
| Faction                      | §9.112        |
| FireAttack                   | §6.54         |
| FireFactor                   | §6.11         |
| FireFactorRow                | §CRT          |
| FoKVictoryLevel              | §9.35         |
| Fort                         | §5.25, §6.54  |
| Friendlies                   | §6.52         |
| FriendliesTransport          | §5.21         |
| GameState                    | §4            |
| GameTime                     | §9.12         |
| GameTurnIndex                | §4            |
| GunboatId                    | §2.32, §6.64  |

|                        |               |
|------------------------|---------------|
| GunboatMovement        | §5.24         |
| GunboatStack           | §5.51         |
| HISTORICAL_TURN_TRACK  | §9.22         |
| HexDirection           | §5.11         |
| HexDistance            | §6.22, §7.5   |
| HexsideKind            | §5.23         |
| HexsideRef             | §5.23         |
| HistoricalVictoryLevel | §9.24         |
| Howitzer               | §2.31         |
| HowitzerFire           | §6.64         |
| HowitzerResolution     | §6.64         |
| Immobile               | §5.25         |
| Khor                   | §5.44         |
| Location               | §9.112        |
| LosFirerTerrain        | §6.3          |
| LosResult              | §6.3          |
| LosSpecialNote         | §6.3          |
| LosTargetTerrain       | §6.3          |
| MaximSecondAndHowitzer | §6.42         |
| MeleeAttack            | §7.7          |
| MeleeCombat            | §7.3          |
| MeleeFactor            | §7.1          |
| MeleeModifier          | §7.7          |
| MineResult             | §10.1         |
| MovementAllowance      | §5.11         |
| MovementPoints         | §5.11         |
| NamedGunboat           | §2.32         |
| NileFlow               | §5.11, §5.24  |
| Old                    | §2.32         |
| OldGunboat             | §2.32         |
| OptionalRule           | §10           |
| OverLimit              | §5.51         |
| PendingMelee           | §4            |
| Phase                  | §4            |
| PlaceReinforcements    | §9.112        |
| Range                  | §6.22         |
| RangeBand              | §6.16, §6.22  |
| RetreatBeforeMelee     | §7.5          |
| RiverMine              | §10.1, §10.12 |
| RoyalEngineers         | §6.53         |
| ScatterDirection       | §6.64         |
| SetupLetter            | §9.112        |
| SpriteAnnotation       | §2.3          |
| Terrain                | §6.23         |



|                              |                               |
|------------------------------|-------------------------------|
| TurnEntry                    | §9.12                         |
| TurnEvent                    | §8.2                          |
| TurnLabel                    | §9.12                         |
| UnitFormKind                 | §2.3                          |
| UnitKind                     | §7.4                          |
| UnitMovement                 | §5.11                         |
| UnitProfile                  | §2.3                          |
| UnitState                    | §6.54                         |
| VictoryLedger                | §9.14                         |
| VictoryPoints                | §9.14                         |
| VpEvent                      | §9.14                         |
| VpSource                     | §9.14                         |
| Wall                         | §5.44                         |
| WeaponClass                  | §2.31, §6.6, §6.61, §6.62     |
| WhiteNileMouth               | §9.345                        |
| Zariba                       | §5.44                         |
| ZaribaThornHedge             | §9.23, §9.231                 |
| ZaribaTrench                 | §9.23, §9.232                 |
| ZaribaTrenchEntrenched       | §9.23, §9.232                 |
| ZocReason                    | §5.41, §5.44, §6.54           |
| advance_phase                | §4                            |
| ae_range_effects             | §6.22                         |
| apply_advance_after_combat   | §6.82, §7.6                   |
| apply_construct_zariba       | §5.3                          |
| apply_demolition             | §6.53                         |
| apply_friendlylies_transport | §5.21                         |
| apply_howitzer_fire          | §6.64                         |
| apply_melee_combat           | §7.3                          |
| apply_place_reinforcements   | §9.112                        |
| apply_retreat_before_melee   | §7.5                          |
| apply_river_mine             | §10.1, §10.12, §10.13, §10.14 |
| apply_sink_chain             | §10.23                        |
| blocks_advance_after_combat  | §6.82                         |
| blocks_los                   | §6.23, §6.3                   |
| blocks_melee                 | §7.2                          |
| blocks_movement              | §5.23                         |
| brigade_integrity            | §5.54                         |
| can_advance_after_combat     | §6.7, §6.82, §7.6             |
| can_fire_at                  | §6.64                         |
| can_melee                    | §7.2, §7.4                    |
| can_move_gunboat             | §10.22                        |
| can_move_unit                | §4                            |
| can_move_unit_to             | §5.22, §5.26, §5.43           |
| can_retreat_before_melee     | §5.23, §7.5                   |

|                             |              |
|-----------------------------|--------------|
| check_gordon_palace         | §9.346       |
| combat_results_table        | §CRT         |
| constructing_zariba         | §5.3         |
| defense_modifier            | §6.23        |
| demolishing                 | §6.53        |
| dervish_range_effects       | §6.22        |
| die_modifier                | §6.24        |
| effective_movement_at_night | §8.1         |
| effective_range_at_night    | §8.1         |
| end_player_turn             | §4, §5.13    |
| fires_twice                 | §6.42        |
| first_player                | §9.211       |
| from_superiority            | §9.14        |
| from_total                  | §CRT         |
| halve                       | §8.1         |
| has_combat_factors          | §6.51        |
| hex_has_enemy_fort          | §9.344       |
| hex_in_enemy_zoc            | §4, §5.43    |
| hit_target_hex              | §6.64        |
| howitzer_scatter            | §6.64        |
| in_deployment_zone          | §9.212       |
| is_boat                     | §5.24        |
| is_crossroad                | §5.11        |
| is_friendlylies             | §5.21, §6.52 |
| is_gordon                   | §9.346       |
| is_los_trees                | §6.23        |
| is_nile_mouth_crossing      | §9.345       |
| loaded_on                   | §5.21        |
| los_table                   | §6.21, §6.3  |
| may_act                     | §6.54        |
| may_attack_this_turn        | §5.3         |
| may_be_melee_attacked       | §7.1         |
| may_melee_attack            | §7.4         |
| may_retreat_before_melee    | §7.5         |
| movement_cost               | §5.11        |
| movement_cost_with_road     | §5.11        |
| mp_spent                    | §5.12        |
| net_modifier                | §6.54        |
| new                         | §4           |
| overlay_color               | §5.52        |
| passable_by_land            | §5.11        |
| points                      | §9.14        |
| range_band_for              | §9.343       |
| roads                       | §5.11        |

|                       |              |
|-----------------------|--------------|
| score_elimination     | §9.14        |
| sum                   | §7.1         |
| sum_to_row            | §6.14        |
| superiority           | §9.14        |
| terrain_effects_chart | §5.11        |
| total_for             | §9.14        |
| turn_marker_pixel     | §9.12        |
| unit_projects_zoc     | §5.41, §5.44 |
| value                 | §4           |
| who_scores            | §9.14        |